From Global Search to Local Lookup: Scalable Knowledge Base Completion via Differentiable Guarded Logic

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Abstract

State-of-the-art Knowledge Base Completion (KBC) models, ranging from geometric embeddings to Graph Neural Networks (GNNs), achieve impressive predictive performance but share a critical limitation: they act as black boxes that often ignore data consistency, producing predictions that violate rigorous ontological constraints. While Neural-Symbolic (NeSy) frameworks attempt to bridge this gap by integrating logical rules, they face a prohibitive scalability barrier. Existing approaches rely on unguarded universal quantification that demands global search, causing combinatorial explosions and execution failures on large-scale knowledge bases.

In this paper, we propose **GuardNet**, a differentiable reasoning framework designed to break this expressivity-scalability deadlock. Our core contribution is leveraging the Guarded Fragment (GF) of first-order logic to fundamentally restructure the computational graph of reasoning. We demonstrate that the GF's syntactic "guard" acts as a topological constraint that transforms intractable global quantification into efficient neighborhood-restricted lookups that strictly align logical evaluation with the underlying graph structure. For sparse knowledge bases, this paradigm shift eliminates the quadratic grounding overhead that plagues traditional NeSy systems, reducing complexity to linear in the number of edges without sacrificing the semantic expressiveness required for complex reasoning. Experiments on massive benchmarks show that GuardNet is the first NeSy framework to scale to such large knowledge bases, succeeding where NeSy and probabilistic baselines fail to converge within 72 hours, while significantly outperforming SOTA geometric embedding and GNN models in both accuracy and consistency.

The source code, test data, alongside the extended experimental results are available at https://anonymous.4open.science/r/guardnet/ to support reproducibility and future research.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**; **Neural networks**; **Machine learning**; *Learning latent representations*; • **Information systems** → *Graph-based database models.*

Keywords

Neuro-Symbolic AI, Knowledge Base Completion, Guarded Fragment, Differentiable Fuzzy Logic, Scalable Reasoning

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1 Introduction

Knowledge Base Completion (KBC) is a fundamental task in data management, requiring systems to infer missing information from large-scale, incomplete knowledge bases [2, 11, 21, 23, 30, 31]. In the Description Logic (DL) context, a knowledge base $\mathcal{K} = (\mathcal{T}, \mathcal{A})$ is typically defined as the union of a TBox (aka ontology) $\mathcal{T}$ (terminological axioms defining concept hierarchies and relationships) and an ABox $\mathcal{A}$ (assertional facts about individuals). Equivalently, from a knowledge graph perspective, the TBox corresponds to the ontological schema while the ABox corresponds to the instance-level graph of entities and relations. Modern biomedical ontologies such as SNOMED CT [48] (377K concepts) and Gene Ontology [4] (44K concepts; abbreviation: GO) encode rich taxonomic hierarchies in their TBoxes, while protein interaction networks demand reasoning over millions of relational facts in their ABoxes [52]. These applications require *multi-hop logical inference*—deducing that a drug treats a disease by composing intermediate relationships—rather than shallow pattern matching [22, 28, 36, 41, 42, 50, 70].

Prominent approaches to KBC fall into two camps, each with fundamental limitations. *Geometric embedding methods* [11, 32, 51, 56, 64, 69] and their ontology-aware extensions [1, 24, 27, 66, 72] scale linearly but sacrifice logical fidelity, learning implicit patterns rather than explicit rules. *Neural-Symbolic (NeSy) frameworks* [7, 14, 44, 58] preserve logical semantics but suffer catastrophic scalability failures on real-world knowledge bases. This creates an **expressivity–scalability dilemma** that has stymied progress toward production-grade logical reasoning systems [15, 37].

The Symbol Grounding Bottleneck. We identify the root cause of this dilemma as the *grounding bottleneck* inherent in existing NeSy architectures, as illustrated in **Fig. 1**. Consider evaluating a universally quantified formula $\forall x, y.\, R(x, y) \rightarrow S(x, y)$ over a domain of N entities. Standard frameworks such as LTN [7] and Neural LP [65] implement this via *dense tensor broadcasting*: they materialize the full Cartesian product $N \times N$, allocating $O(N^2)$ memory to represent all possible variable bindings. For SNOMED CT with $N = 377$K, this requires 1.4×10^{11} tensor elements—far exceeding any GPU's capacity.

This is analogous to executing a nested-loop join without indices in a database system. The vast majority of computed entries correspond to *vacuous truths*—pairs (x, y) where $R(x, y)$ is false, making the implication trivially satisfied. These computations are semantically meaningless yet consume identical resources to genuine inferences, creating massive waste that triggers out-of-memory (OOM) failures at modest scales. As we demonstrate empirically, LTN and Neural LP crash at merely 5K entities, while production knowledge bases contain two orders of magnitude more.

LTN does offer a mechanism called *guarded quantifier* [7] that applies a Boolean mask to exclude certain variable bindings from aggregation. However, this mask is evaluated at *runtime*: the full N^2 domain is materialized first, each pair is tested against the mask

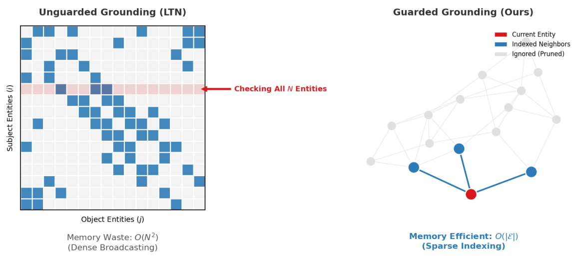

Figure 1: The Grounding Bottleneck. Left (Unguarded): Standard NeSy approaches materialize the full $N \times N$ Cartesian product, with gray cells representing vacuous truths that consume $O(N^2)$ memory. Right (Guarded): GuardNet uses guard atoms as topological indices, restricting computation to the guard extension $\mathrm{ext}(\alpha)$.

condition, and only then are non-qualifying pairs discarded. The mask evaluation itself remains $O(N^2)$, so no asymptotic complexity reduction is achieved. This raises a deeper question: can the restriction be imposed at *formula-writing time*, so that the grounding scope is determined by the syntactic structure of the logic before any computation begins?

The Landscape of Existing Solutions. The field of NeSy AI has developed three major paradigms to integrate neural learning with symbolic reasoning [10, 16], each representing a different trade-off:

- *Probabilistic Logic Programming.* Systems such as ProbLog [40], DeepProbLog [34], NeurASP [67], DeepStochLog [62], and NeuPSL [39] integrate neural networks with probabilistic inference. While principled for handling uncertainty, their reliance on explicit symbolic grounding limits scalability to knowledge bases with tens of thousands of facts.
- *Logic as Soft Constraint.* Another paradigm treats logical axioms as regularization terms during training. This includes Semantic-Based Regularization [17], Semantic Loss [63], Deep Logic Models [38], and differentiable SAT solvers [61]. These methods achieve efficiency by relaxing logical strictness, but struggle to enforce complex multi-hop dependencies.
- *Differentiable Fuzzy Logic.* Methods including DFL [58], LTN [7], logLTN [8], Neural Theorem Provers [44], TensorLog [14], and semantic entailment systems [53, 54] embed first-order logic (FOL) into continuous vector spaces via fuzzy semantics [25]. While maximally expressive, they inherit FOL's undecidability [13, 57] and the quadratic grounding bottleneck.

Orthogonally, GNN-based approaches [55, 59, 60, 71, 73] achieve scalability through implicit message passing, but lack explicit logical semantics and cannot express negation, disjunction, or universal quantification [29]. Recent advances in temporal reasoning [68], ontology matching [18], and KG population [31] further highlight the need for scalable yet logically grounded systems.

Guard-based Tractability. The idea of restricting a logical language to achieve tractable differentiable reasoning has precedent: TensorLog [14] identifies polytree-limited SDKGs as a tractable fragment with linear-time inference, targeting the #P-hardness of probabilistic inference. However, TensorLog's restriction is orthogonal to ours: it constrains the variable-sharing structure of clauses (polytree) to avoid exponential proof paths, whereas GuardNet constrains the quantifier scope via a guard atom to reduce *grounding size* from $O(N^2)$ to $O(|\mathcal{E}|)$. More broadly, several NeSy systems restrict quantification in practice (e.g., LTN's runtime mask, Neural-LP's sparse-matrix multiplication, NTP's backward chaining against observed facts, PSL/NeuPSL's typed domains, pLogicNet's variational selection), but none provides a provable grounding-size bound from a named logical fragment. In database theory, GTGDs [12] use a syntactic guard condition to confine the chase to existing tuples, achieving PTIME data complexity for atomic queries. Vadalog [9] operationalizes a generalization of this idea (Warded Datalog$^{\pm}$, strictly more expressive than Guarded Datalog$^{\pm}$) for enterprise-scale reasoning. GuardNet is the first system to embed this guard-based tractability mechanism as a differentiable inductive bias, connecting NeSy grounding to the DB-theoretic tractability of GTGDs.

Our Insight: The Guarded Fragment as Computational Primitive. We answer this question affirmatively. Building on the GTGD tractability framework discussed above, we identify the GF guard condition as the missing inductive bias for scalable neural-symbolic reasoning. Concretely, we build on the **Guarded Fragment (GF)** of FOL [3], the logical foundation underlying GTGDs. In GF, every quantified variable must appear within a *guard atom*—a relational atom that restricts the variable's scope to a local neighborhood. Unlike LTN's runtime mask, the GF guard condition $\mathrm{FV}(\psi) \cup \bar{x} \subseteq \mathrm{Vars}(\alpha)$ is a *syntactic* property of the formula itself: it determines the grounding scope at formula-writing time, before any tensor is allocated.

The key insight is that the guard atom functions as a topological index, the differentiable analogue of a database index that converts a nested-loop join into a direct lookup. Rather than broadcasting over the entire domain, quantification is restricted to entities connected by existing edges in the knowledge base. For a guarded formula $\forall \bar{x}(\alpha(\bar{x}) \rightarrow \psi)$, this transforms the enumeration over all $N^{|\bar{x}|}$ possible tuples into a sparse traversal over only those tuples that actually satisfy the guard α in the knowledge base. In sparse knowledge bases—the predominant structure in real-world applications where the number of edges $|\mathcal{E}| \ll N^2$—this yields substantial complexity reductions. Crucially, GF is not a toy fragment: it subsumes $\mathcal{ALC}$ [47] and covers nearly all constructs of $\mathcal{EL}^{++}$ [5] (the only exceptions are role composition and transitivity; see §5), while remaining decidable [6, 20]. This means the vast repository of industrial ontologies and knowledge bases can be directly translated into GF without loss of semantics, enabling rigorous logical reasoning at unprecedented scale.

The GuardNet Framework. We operationalize this insight in GuardNet, the first end-to-end differentiable framework that leverages the GF as a structured inductive bias for neural reasoning over knowledge bases. Our contributions are:

- **Principled Fuzzy Semantics (§3).** Building on theoretical foundations of fuzzy DLs [49] and differentiable fuzzy operators [58], we present a semantics grounded in the Product t-norm with a Sigmoidal Reichenbach implication. This combination ensures smooth, non-vanishing gradients that prevent dead zones plaguing existing systems.

- **Guarded Grounding with Linear Complexity (§4.1).** We develop a novel symbol grounding mechanism that uses guard atoms as topological indices, transforming quantified formulas into sparse, neighborhood-restricted tensor operations. This reduces memory complexity from $O(N^2)$ to $O(|\mathcal{E}|)$, enabling—for the first time—rigorous FOL-based reasoning on knowledge bases with hundreds of thousands of concepts. This neuralizes the tractability mechanism of GTGDs [12], establishing a formal bridge between database-theoretic query answering and differentiable neural reasoning. Unlike parallel embedding training approaches [26] that distribute existing algorithms, our method fundamentally restructures the computation itself.
- **Hybrid Domain Strategy (§4.2, 4.3, 4.4)** We propose a novel witness construction mechanism that systematically prevents the model from satisfying axioms vacuously. This addresses a fundamental learning pathology—where models learn trivial "all-false" solutions—ensuring that logical satisfaction reflects genuine semantic understanding.

Empirical Validation. We conduct extensive experiments on four challenging benchmarks. GuardNet consistently outperforms 15 state-of-the-art baselines spanning five paradigms. Notably, while SOTA NeSy methods fail with OOM at 5K/10K entities, GuardNet scales linearly to 377K concepts—a **75× improvement in scalability** with simultaneous gains in accuracy. Our ablation studies further validate the critical role of the **Guarded Grounding** and **Hybrid Domain** strategies in achieving these results.

2 Preliminaries: Guarded Fragment

In this section, we introduce the **GF** [3], a decidable fragment of FOL that strikes a balance between expressivity and decidability & complexity.

Let $\mathcal{P}$, $\mathcal{C}$, and $\mathcal{V}$ be countably infinite and pairwise disjoint sets of **predicate**, **constant**, and **variable** symbols, respectively. A **signature** $\Sigma = (\mathcal{P}, \mathcal{C})$ specifies the non-logical vocabulary used to construct well-formed GF formulas. Following the standard definition of GF [3], **terms** are restricted to constants and variables only, excluding function symbols present in full FOL.

An **atom** has the form $P(t_1, \ldots, t_n)$, where $P \in \mathcal{P}$ is an n-ary predicate and $t_1, \ldots, t_n$ are terms. For any atom $\alpha = P(t_1, \ldots, t_n)$, we define $\mathrm{Vars}(\alpha) = \{t_i \mid t_i \in \mathcal{V}\}$ as the set of variables occurring in α. In a quantified formula $\forall x\, \phi / \exists x\, \phi$, we say the quantifier **binds** the variable x, and ϕ is the **scope** of the quantifier. An **occurrence** of a variable x in a formula ϕ is **bound** if it lies within the scope of a quantifier that binds x; otherwise, the occurrence is **free**. A variable x is a **free variable** of ϕ if x has at least one free occurrence in ϕ. We denote by $\mathrm{FV}(\phi)$ the set of all free variables of ϕ. A GF formula with no free variables is called a **sentence**.

Definition 1 (Syntax of GF). *GF formulas are defined inductively as follows:*

- *Every atom is a GF formula;*
- *If ϕ and ψ are GF formulas, then so are $\neg\phi$, $\phi \wedge \psi$, $\phi \vee \psi$, and $\phi \to \psi$;*
- *If ψ is a GF formula, $\bar{x}$ is a tuple of variables, and α is an atom satisfying the **guard condition**:*

$$\mathrm{FV}(\psi) \cup \bar{x} \subseteq \mathrm{Vars}(\alpha),$$

then $\exists\bar{x}\,(\alpha \wedge \psi)$ and $\forall\bar{x}\,(\alpha \to \psi)$ are GF formulas.

In guarded quantification, the atom α serves as a **guard** for the **body** ψ. The guard condition requires that all variables involved in the quantification—both the free variables of the body and the quantified variables—appear in the guard. This fundamentally relativizes quantification: instead of ranging over the entire domain, quantified variables are restricted to the local neighborhood defined by the guard. For instance, $\exists x\,(R(x, y) \wedge \psi(x, y))$ restricts x to objects that are R-related to y, effectively performing a one-hop neighbor lookup on a KB, directly mirroring the message-passing paradigm in GNNs. We illustrate with two axioms from SNOMED CT, which serve as running examples throughout the paper.

▹ Example (Running)

Consider the SNOMED CT axiom:

$$\mathsf{Pneumonia} \sqsubseteq \mathsf{LungDisease}$$

stating that every pneumonia is a lung disease. Its GF translation is:

$$\forall x\,(\mathsf{Pneumonia}(x) \to \mathsf{LungDisease}(x))$$

Here the unary atom $\mathsf{Pneumonia}(x)$ is the guard. The guard extension $\mathrm{ext}(\mathsf{Pneumonia})$ contains only the entities asserted or inferred to be pneumonia instances. In SNOMED CT (N=377K concepts), this might be a few hundred entities. An unguarded system evaluates the implication for all 377K entities; GuardNet evaluates it only over $\mathrm{ext}(\mathsf{Pneumonia})$.
Now consider a richer axiom:

$$\mathsf{Pneumonia} \sqsubseteq \exists\mathsf{hasFinding}.\mathsf{Fever}$$

stating that every pneumonia has fever as a clinical finding. Its GF translation is:

$$\forall x\,(\mathsf{Pneumonia}(x) \to \exists y\,(\mathsf{hasFinding}(x, y) \wedge \mathsf{Fever}(y)))$$

The guard $\mathsf{Pneumonia}(x)$ restricts the outer quantification, and $\mathsf{hasFinding}(x, y)$ guards the inner existential. Together, they confine grounding to the two-hop neighborhood of pneumonia entities in the knowledge graph, yielding $O(|\mathcal{E}|)$ instead of $O(N^2)$.

Based on this syntax, we define the central data structure used in our framework. A **knowledge base** (KB) $\mathcal{K}$ is a finite set of GF sentences constructed over the signature Σ. Analogous to the terminology in DLs, we partition $\mathcal{K}$ into an ABox $\mathcal{A}$, consisting of ground facts (i.e., variable-free atoms), and a TBox $\mathcal{T}$, consisting of quantified formulas that describe general terminological knowledge. The **size** of the KB, denoted $|\mathcal{K}|$, is defined as the total number of symbols required to write down all formulas in $\mathcal{K}$.

GF inherits its semantics from standard FOL: a structure $\mathfrak{A} = \langle \Delta, \cdot^{\mathfrak{A}} \rangle$ assigns a domain Δ, interprets constants as domain elements and predicates as relations over Δ, and evaluates satisfaction inductively in the usual way. The only difference from unrestricted FOL

Table 1: Fuzzy operators selected for GuardNet (bold) and alternatives evaluated. Full gradient analysis in App. A.3.

Category	Operator	Gradient note
T-norm ($\wedge$)	**Product (ours)**	Adaptive: $\partial T/\partial a{=}b$
	Gödel	Winner-take-all
	Łukasiewicz	Dead zone when $a{+}b{\le}1$
	Yager (p=2)	Tunable interpolation
	Hamacher	Coupled sensitivity
Implication ($\rightarrow$)	**Sigmoidal Reichenbach**	Boundary-focused; no plateaus
	Standard Reichenbach	Linear; loss = $a(1{-}c)$
	Goguen	Zero plateau ($a{\le}c$)
Quantifier ($\forall/\exists$)	**Generalized Mean**	Tunable; $[0,1]$-bounded

is that quantifiers in GF are syntactically constrained by guards; the semantic evaluation rules for $\exists\bar{x}\,(\alpha \wedge \psi)$ and $\forall\bar{x}\,(\alpha \rightarrow \psi)$ are identical to their FOL counterparts. We provide the full formal definition in App. A.2. GF is decidable [3], and enjoys the finite model property [20], i.e., every satisfiable GF formula has a model of a finite domain. These properties make GF a fragment where expressive logical reasoning remains computationally feasible. From a database-theoretic perspective, the guard condition in Def. 1 is precisely the syntactic constraint underlying GTGDs [12]. GuardNet operationalizes this classical mechanism in a differentiable neural architecture, as detailed in §4.1.

3 A Differentiable Semantics for Learning

Classical GF provides decidable reasoning [3], yet real-world applications demand the ability to handle the uncertainty and vagueness inherent in noisy data. To address this, we extend GF with fuzzy semantics, mapping atomic formulas like Fever(x) to continuous truth values in $[0,1]$. However, the choice of fuzzy operators is a critical hyperparameter defining the **geometry of the optimization landscape** [58]. A poor choice can lead to vanishing gradients or stagnation. Tab. 1 summarizes the operator configuration used by GuardNet; a comprehensive comparison of all candidate operators with gradient analysis is provided in App. A.3.

3.1 Fuzzy Operator Selection Rationale

We select the **Product** t-norm because it provides adaptive gradient scaling: $\partial T/\partial a = b$, so each input receives signal proportional to its partner's confidence, balancing bottleneck correction with holistic learning. Alternatives suffer from dead zones (Łukasiewicz), winner-take-all sparsity (Gödel), or premature fixation (Hamacher). The **Probabilistic Sum**, as the Product dual, inherits these balanced properties for disjunction. For implication, the **Sigmoidal Reichenbach** [58] concentrates gradient mass on decision boundaries, addressing the zero-plateau problem of R-implications (Goguen) and the linear treatment of all violations by the standard Reichenbach S-implication. A full gradient analysis across all five t-norm families, three implication types, and three aggregation strategies is provided in App. A.3.

3.2 Guarded Quantifiers as Parameterized Local Pooling

In GF, quantification is always restricted to a local neighborhood defined by the guard atom. To support our framework's logic-agnostic design—where operators may range from strict (Gödel) to smooth (Product)—the aggregation mechanism must be equally adaptable. We adopt the **Generalized Mean (p-mean)** as a universal, parameterizable pooling operator. By adjusting the exponent $p \in [1,\infty)$, it provides a continuous spectrum: reducing to the arithmetic mean at $p = 1$ (matching Łukasiewicz linearity) and converging to the hard extremum as $p \rightarrow \infty$ (matching Gödel). Let $\mathbf{z} = (z_1, \ldots, z_k)$ be the truth values evaluated over k guard-satisfying neighbors.

Existential quantifier ($\exists$). We employ the standard Generalized Mean to aggregate evidence for satisfaction:

$$\mathcal{A}^{\exists}(\mathbf{z}; p) = \left(\frac{1}{k}\sum_{i=1}^{k} z_i^p\right)^{\frac{1}{p}} \tag{1}$$

Higher p focuses on the best evidence (approaching max); lower p aggregates uniformly.

Universal quantifier ($\forall$). We employ the **Generalized Mean Error** to aggregate deviation from perfect truth [7]:

$$\mathcal{A}^{\forall}(\mathbf{z}; p) = 1 - \left(\frac{1}{k}\sum_{i=1}^{k} (1 - z_i)^p\right)^{\frac{1}{p}} \tag{2}$$

This formulation prevents gradient vanishing when the majority of neighbors satisfy the condition. As $p \rightarrow \infty$, the result converges to the minimum (the "weakest link"). Since each $z_i \in [0,1]$, both $\mathcal{A}^{\exists}$ and $\mathcal{A}^{\forall}$ are guaranteed to remain in $[0,1]$ for all $p \ge 1$, unlike unbounded approximations such as LogSumExp.

Numerical stability. As noted by Badreddine et al. [7], the gradient of the p-mean can explode when inputs approach boundaries (0 or 1) for large p. To ensure stability across arbitrary operator combinations, we implement the *Stable Product* strategy, applying a projection $\pi(x) = (1-\epsilon)x + \epsilon$ to inputs before aggregation.

3.3 Generalized Fuzzy Satisfaction

Finally, we integrate the operator choices discussed above into a unified semantic framework. GuardNet treats the logical semantics as a hyperparameter, allowing the system to be configured based on domain constraints (e.g., noise levels, graph sparsity). We define the generalized fuzzy satisfaction degree $[\![\phi]\!]_{\mathfrak{A},\rho} \in [0,1]$ as follows.

Definition 2 (Generalized Fuzzy Satisfaction). *Let $\mathfrak{A}$ be a fuzzy structure, ρ a variable assignment, and $O = \{T, S, I, \mathcal{A}_p^{\exists}, \mathcal{A}_p^{\forall}\}$ be a selected set of differentiable fuzzy operators parameterized by p. The satisfaction degree is defined recursively:*

$$[\![P(\bar{t})]\!]_{\mathfrak{A},\rho} = P^{\mathfrak{A}}(\bar{t}^{\mathfrak{A},\rho}) \quad [\![\neg\phi]\!]_{\mathfrak{A},\rho} = 1 - [\![\phi]\!]_{\mathfrak{A},\rho}$$
$$[\![\phi \wedge \psi]\!]_{\mathfrak{A},\rho} = T([\![\phi]\!]_{\mathfrak{A},\rho}, [\![\psi]\!]_{\mathfrak{A},\rho})$$
$$[\![\phi \vee \psi]\!]_{\mathfrak{A},\rho} = S([\![\phi]\!]_{\mathfrak{A},\rho}, [\![\psi]\!]_{\mathfrak{A},\rho})$$
$$[\![\exists\bar{x}\,(\alpha \wedge \psi)]\!]_{\mathfrak{A},\rho} = \mathcal{A}_p^{\exists}\left(\left\{[\![\alpha \wedge \psi]\!]_{\mathfrak{A},\rho'} \mid \bar{a} \in \Delta^{|\bar{x}|}\right\}\right)$$
$$[\![\forall\bar{x}\,(\alpha \rightarrow \psi)]\!]_{\mathfrak{A},\rho} = \mathcal{A}_p^{\forall}\left(\left\{[\![\alpha \rightarrow \psi]\!]_{\mathfrak{A},\rho'} \mid \bar{a} \in \Delta^{|\bar{x}|}\right\}\right)$$

where $\rho' = \rho[\bar{x} \mapsto \bar{a}]$ denotes the assignment update.

This formulation provides GuardNet with continuous adaptability: it can interpolate between **strict, rule-bound logic** (using Gödel operators and large p) for precise reasoning in clean domains, and **probabilistic, soft reasoning** (using Product operators and small p) for robust learning in noisy, uncertain KBs.

4 A Differentiable Fuzzy GF Framework

To operationalize the theoretical advantages of GF for learning, we present GuardNet, a NeSy framework built upon three core components: (1) a scalable neural grounding mechanism; (2) a unified fuzzy loss function; and (3) a semantically-aware domain construction strategy. The overall architecture is illustrated in **Fig. 2**.

4.1 Neural Grounding Mechanism

The bridge between logic and neural networks is the **grounding** function $\mathcal{G}_\theta$. Before detailing our implementation, we formally analyze the bottleneck that necessitates our architectural departure from standard neural-symbolic approaches.

4.1.1 The Bottleneck: Unguarded Grounding and Gradient Dilution. Standard approaches (e.g., LTN) typically ground FOL rules by instantiating variables over the entire domain Δ. For a rule like $\forall x, y\ (R(x, y) \rightarrow C(y))$, standard semantics treat quantifiers as **unguarded**, aggregating over the full Cartesian product:

$$[\![\forall x, y\ \phi(x, y)]\!] = \bigotimes_{u,v \in \Delta} [\![\phi(u, v)]\!] \tag{3}$$

This formulation introduces two key failures in large-scale learning:

- **System Implementation Bottleneck:** Physically evaluating Eq. 3 in deep learning frameworks requires **tensor broadcasting** to materialize all variable combinations. For a domain size of $|\Delta| \approx 3.7\times10^5$ (e.g., SNOMED CT), constructing a dense pairwise tensor of shape $[|\Delta|, |\Delta|, d]$ attempts to allocate hundreds of gigabytes of GPU memory, triggering immediate OOM errors.
- **Optimization Bottleneck (Gradient Dilution):** Even if memory permits, existing frameworks remain agnostic to the sparsity of R. They evaluate the implication $R(u, v) \rightarrow C(v)$ for *every* pair. In sparse KBs, $R(u, v)$ is false for the vast majority of pairs. Since the logical implication *False* $\rightarrow \ldots$ trivially evaluates to True (1), the model wastes $O(|\Delta|^2)$ computation verifying these tautologies. Crucially, this creates a gradient dilution problem: the learning signal from the few actual edges (where R is true) is mathematically drowned out by millions of vacuously satisfied instances, stalling convergence.

4.1.2 The Solution: Guarded Sparse Indexing. GuardNet resolves this by strictly enforcing the **Guarded Grounding** paradigm. Instead of broadcasting over the domain, we leverage the GF syntax to treat the guard atom α as a topological index.

Complexity Analysis. Let $\phi = \forall\bar{x}(\alpha(\bar{x}) \rightarrow \psi(\bar{x}))$ be a guarded axiom where α is the guard atom and $|\bar{x}| = k$ is the number of quantified variables. We define the *guard extension* as:

$$\text{ext}(\alpha) = \{(a_1, \ldots, a_k) \mid \alpha(a_1, \ldots, a_k) \text{ holds in } \mathcal{K}\}$$

- **Unguarded Grounding:** Standard approaches enumerate all N^k variable assignments (where $N = |\Delta|$ is the domain size), yielding $O(N^k)$ complexity per axiom. For binary predicates ($k = 2$), this is $O(N^2)$.
- **Guarded Grounding:** Our approach enumerates only tuples in $\text{ext}(\alpha)$, yielding $O(|\text{ext}(\alpha)|)$ complexity per axiom.

For a KB $\mathcal{K}$ with m axioms, the total grounding complexity per epoch is:

$$O\left(\sum_{i=1}^{m} |\text{ext}(\alpha_i)|\right) \leq O(m \cdot |\mathcal{E}|_{\max})$$

where $|\mathcal{E}|_{\max} = \max_i |\text{ext}(\alpha_i)|$ is the size of the largest guard extension in $\mathcal{K}$.

Sparsity Assumption. This complexity advantage is predicated on the *sparsity* of real-world KBs. When guard predicates have bounded extension size (i.e., $|\mathcal{E}|_{\max} = O(N)$), guarded grounding achieves complexity linear in the domain size. We note three caveats:

- In dense KBs where guard extensions approach $\Theta(N^2)$, guarded and unguarded grounding have equivalent asymptotic complexity.
- Our experimental datasets (SNOMED CT, GO, PPI networks) exhibit sparse structure with average degree $\bar{d} \ll N$, empirically validating this assumption.
- Even in scale-free networks with high-degree hub nodes, the worst-case guard extension $|\text{ext}(\alpha)|_{\max}$ remains far below N^2. In our densest experimental dataset (N=6,438, $\bar{d}$=34.3; see §5), power-law degree analysis estimates $d_{\max} \approx 300$, so even the largest hub processes $d_{\max}^2/N^2 < 0.3\%$ of the pairs an unguarded approach would evaluate. Hub load imbalance is a scheduling concern, not an asymptotic one; it is handled by our mini-batch strategy which distributes guard extensions across batches.

▹ Example (Guarded Grounding Trace)

Consider a 3-entity subgraph of SNOMED CT with entities $\{\text{pneu}_{01}, \text{bronch}_{02}, \text{fever}_{03}\}$ and a single edge:

$$\text{hasFinding}(\text{pneu}_{01},\ \text{fever}_{03})$$

For the axiom:

$$\forall x\, (\text{Pneumonia}(x) \rightarrow \exists y\, (\text{hasFinding}(x, y) \land \text{Fever}(y)))$$

Unguarded (LTN): Enumerate all $3^2 = 9$ pairs for hasFinding, evaluate the implication for each. 8 out of 9 produce vacuous truths (hasFinding $=$ false $\Rightarrow$ implication $= 1$), wasting computation and diluting gradients.
Guarded (GuardNet): Look up:

$$\text{ext}(\text{hasFinding}) = \{(\text{pneu}_{01},\ \text{fever}_{03})\}$$

Evaluate the implication only for this single pair. Total cost: $O(|\mathcal{E}|) = O(1)$ instead of $O(N^2) = O(9)$.

Computational vs. Semantic Properties. The $O(|\mathcal{E}|)$ bound is a *computational* property derived from the syntactic structure of the guard atom and the knowledge graph topology. It does not depend on the choice of fuzzy operators: whether conjunction is interpreted via Product, Gödel, or any other t-norm, the set of tuples

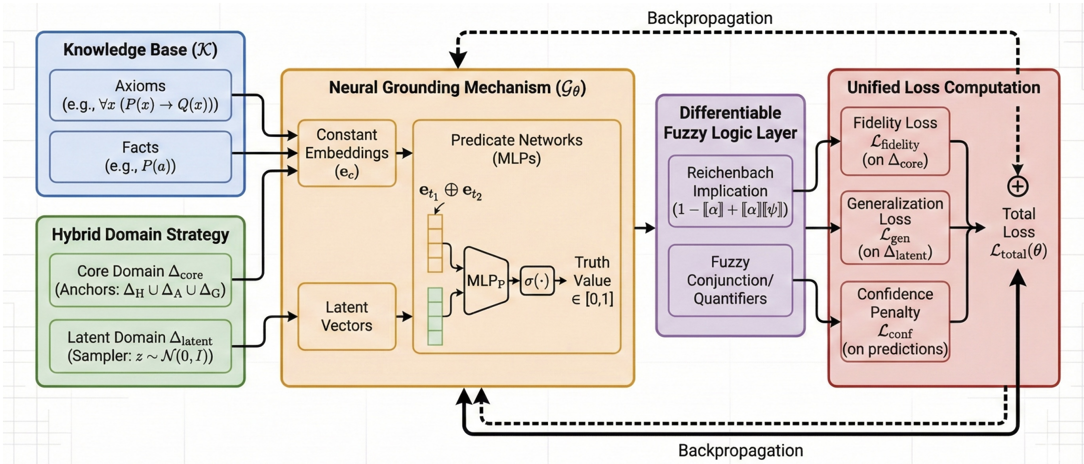

Figure 2: GuardNet framework: Neural Grounding mechanism, Hybrid Domain construction, and unified loss computation.

to be evaluated remains $\text{ext}(\alpha)$. Classical semantic properties of GF (decidability, finite model property) do not automatically transfer to the fuzzy relaxation; the complexity bound, however, does, because it is determined before any truth-value computation begins.

For sparse grounding, GuardNet uses a *closed-world guard index*: for each atomic guard relation α_i, the observed guard extension $\text{ext}_{\mathcal{K}}(\alpha_i)$ is constructed from the KB, and tuples outside this extension are assigned zero guard truth for the purpose of grounding. The following proposition makes explicit that, under this masked grounding semantics, only tuples inside the guard extension can contribute nonzero violation.

Proposition 1 (Soundness of masked guarded evaluation). *Consider a guarded axiom $\phi_i = \forall \bar{x}\,(\alpha_i(\bar{x}) \to \psi_i(\bar{x}))$ where α_i is the atomic guard relation. Under the masked guard-index semantics, $[\![\alpha_i(c)]\!]_{\text{mask}} = 0$ for all $c \notin \text{ext}_{\mathcal{K}}(\alpha_i)$. If the fuzzy implication I satisfies $I(0, t) = 1$ for all $t \in [0, 1]$, then every tuple outside the guard extension is vacuously satisfied:*

$$c \notin \text{ext}_{\mathcal{K}}(\alpha_i) \implies I\big([\![\alpha_i(c)]\!]_{\text{mask}}, [\![\psi_i(c)]\!]\big) = 1.$$

Hence, sparse evaluation over $\text{ext}_{\mathcal{K}}(\alpha_i)$ is violation-equivalent to full-domain evaluation under masked guard-index semantics.

The proof is immediate from $I(0, t) = 1$. This condition is satisfied by the normalized Reichenbach-style implication used in our fuzzy semantics.

Remark. *The closed-world guard index is a computational convention for sparse grounding. It specifies which observed guard tuples participate in the differentiable objective and should not be interpreted as a claim that all unobserved ontology facts are false under classical open-world semantics.*

Below we detail the specific components of this efficient grounding mechanism:

- **Constants as Learnable Embeddings:** Each constant symbol $c \in C$ is grounded as a dense vector embedding $\mathbf{e}_c \in \mathbb{R}^d$. Unlike traditional symbolic reasoning where constants are rigid identifiers, here they act as malleable parameters. This optimization process allows the embeddings to capture the semantic properties of entities specifically as constrained by the logical axioms (e.g., learning that two constants appearing in similar logical contexts have similar vector representations). This design choice parallels knowledge graph embedding methods such as TransE [11] and RotatE [51], but with a key difference: in GuardNet, embeddings are jointly optimized under both relational and logical constraints, so the learned geometry must simultaneously satisfy entity-level facts and universal axioms. We ablate this design in §5.3, comparing learnable embeddings against fixed pretrained alternatives.
- **Predicates as Differentiable Functions:** To ground our n-ary predicates in a way that is both expressive and scalable, we employ Multi-Layer Perceptrons (MLPs) as universal function approximators [46]. This choice provides the flexibility to learn arbitrary non-linear relationships without imposing strong prior assumptions on their geometric structure, which is a crucial feature for a general-purpose GF framework.
  For an n-ary predicate, the input strategy to this MLP is straightforward: it is formed by concatenating the n individual term embeddings [19]. Concatenation is chosen as it is a parameter-free, information-preserving operation that makes no prior assumptions about the relationships between predicate arguments, leaving this task entirely to the learnable layers of the predicate network (see **Fig. 2**). The truth value for an atom $P(t_1, \ldots, t_n)$ is then explicitly given by:

$$[\![P(t_1, \ldots, t_n)]\!]_{\mathcal{G}_\theta} = \sigma\left(\text{MLP}_P(\mathbf{e}_{t_1} \oplus \cdots \oplus \mathbf{e}_{t_n})\right) \tag{4}$$

  where $\mathbf{e}_{t_i} \in \mathbb{R}^d$ is the embedding of term t_i and $\oplus$ denotes concatenation. Our predicate MLPs utilize the ReLU activation function in hidden layers for its robustness against vanishing gradients. The final output layer employs a Sigmoid function $\sigma(\cdot)$ to map the network's logit to a fuzzy truth value in the $[0, 1]$ interval, aligning with a probabilistic interpretation of satisfaction.

4.2 The Challenge: Representational Collapse

A critical vulnerability in neural-symbolic learning is the propensity for models to discover *reasoning shortcuts* [35]—devious paths to minimizing the loss function that bypass meaningful learning. We trace the origin of many such shortcuts to a foundational flaw: an improperly constructed interpretation domain. This flaw permits a catastrophic failure mode we term **representational collapse**. Consider the axiom from Example 1:

$$\forall x\,(\mathrm{Pneumonia}(x) \to \exists y\,(\mathrm{hasFinding}(x,y) \wedge \mathrm{Fever}(y)))$$

Representational collapse manifests in two primary forms, as depicted in **Fig. 3**:

- **Collapse to Emptiness**: The model achieves zero loss by simply learning to predict $[\![\mathrm{Pneumonia}(c)]\!] \approx 0$ for all constants c. In this degenerate solution, the axiom is *vacuously satisfied* (since False $\to$ Anything is True), and the model learns nothing about the actual relationship between pneumonia and fever.
- **Collapse to Ambiguity**: The neural networks learn to output a constant value of 0.5 for all inputs, representing maximum uncertainty. This strategy exploits the properties of certain fuzzy loss landscapes to minimize penalties without the risk of making incorrect decisive predictions.

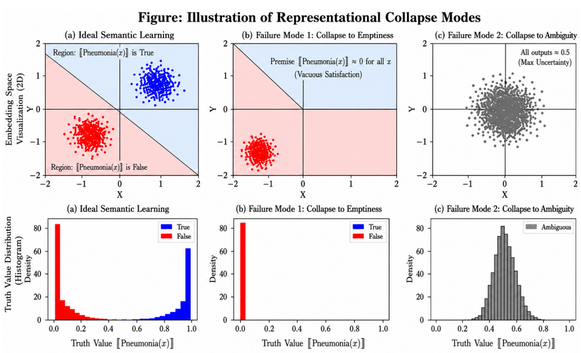

Figure 3: Illustration of Representational Collapse Modes: Collapse to Emptiness (vacuous satisfaction) and Collapse to Ambiguity (maximum uncertainty).

While prior frameworks such as LTN [7] can be effective given dense factual data, real-world ontologies often consist primarily of universal rules (axioms) with sparse factual assertions. This lack of explicit counter-examples makes them highly vulnerable to the aforementioned collapse modes.

Two natural and standard approaches initially seem plausible to address this data scarcity, but they ultimately fail:

- **Failure of Naive Sampling:** First, one might propose relying solely on randomly sampled vectors ("latent constants") from a prior distribution to drive learning. However, when faced with random inputs producing ambiguous outputs (e.g., $[\![\mathrm{Pneumonia}(\mathrm{latent}_1)]\!] \approx 0.5$), the model quickly discovers a trivial optimal strategy: setting $[\![\mathrm{Pneumonia}(x)]\!] \approx 0$ for all inputs. This allows the model to achieve zero loss by making the axiom vacuously satisfied, without learning any semantic structure.
- **Failure of Standard Regularization:** Second, one might employ standard regularization techniques [19] to mitigate collapse. We do actually incorporate such methods: a truth confidence penalty $\mathcal{L}_{\mathrm{conf}} = \sum_i p_i\,(1-p_i)$ that softly encourages predicate outputs p_i to be closer to 0 or 1, along with standard L2 weight decay.

However, these techniques prove fundamentally insufficient. The confidence penalty is "direction-agnostic"—it encourages confident predictions but provides no guidance on what those predictions should be. Even with $\mathcal{L}_{\mathrm{conf}}$, the model can achieve perfect satisfaction by confidently predicting $[\![\mathrm{Pneumonia}(x)]\!] \approx 0$ for all inputs. This simultaneously satisfies the confidence penalty (since 0 is a decisive prediction) and makes the axiom vacuously satisfied (since the premise is never met), while the model learns nothing meaningful about the relationship between pneumonia and fever.

4.3 The Solution: Hybrid Domain Strategy

These failures reveal that robust learning requires explicit semantic anchors that force non-trivial engagement with logical structure. To address this, GuardNet implements a **Hybrid Domain Strategy** that synergistically combines a Core Domain with a Latent Domain, as visualized in **Fig. 4**.

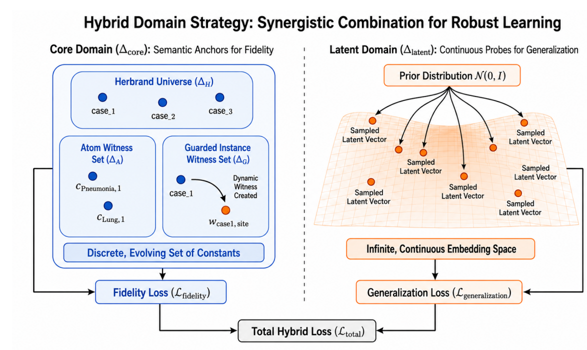

Figure 4: Hybrid Domain Strategy: The Core Domain (Δ_{core}) provides semantic anchors to ensure fidelity; the Latent Domain (Δ_{latent}) drives generalization via continuous sampling.

4.3.1 Core Domain: Ensuring Fidelity via Semantically-Aware Construction.

The Core Domain, Δ_{core}, provides a set of evolving "semantic anchors" that prevent representational collapse through a principled, multi-layered construction process addressing distinct failure modes.

- **Herbrand Domain (Δ_H):** A natural and canonical starting point rooted in classical logic is the **Herbrand Universe** [43]. In the context of a function-free language like GF, this domain consists of all constant symbols that explicitly appear as arguments to predicates within the KB $\mathcal{K}$:

$$\Delta_H = \{c \mid c \in C \text{ and } c \text{ appears as an argument in } \mathcal{K}\} \qquad (5)$$

This domain serves to ground the model's reasoning in the concrete entities for which prior factual knowledge exists. However, relying solely on Δ_H is theoretically insufficient for robust learning. Real-world KBs often consist primarily of universal rules without specific instances, rendering Δ_H sparse or even empty. Consequently, a domain constrained only to these explicit constants remains highly vulnerable to the representational collapse modes described previously.

- **Atom Witness Set (Δ_A):** This component acts as our primary defense against global collapse to emptiness, addressing the foundational problem of predicate existence in the absence of explicit facts. For every n-ary predicate $P \in \mathcal{P}$ that is not currently grounded by factual assertions in $\mathcal{K}$, we introduce a unique tuple of n fresh witness constants, $(c_{P,1}, \ldots, c_{P,n})$. Through dedicated loss terms, we actively enforce a high truth value for these grounded atoms, i.e., $[\![P(c_{P,1}, \ldots, c_{P,n})]\!] \approx 1$. This mechanism guarantees that every concept in the theory—such as Pneumonia($c_{\text{Pneu},1}$) or Fever($c_{\text{Fever},1}$)—is interpreted as a non-empty set from the outset, thereby rendering a total collapse to emptiness impossible.
- **Guarded Instance Witness Set (Δ_G):** While atom witnesses break the initial stalemate of emptiness, they are insufficient on their own. A model might satisfy atom witnesses while still learning only vague statistical associations—maintaining moderate truth values that achieve acceptable loss through quantifier aggregation without learning precise relationships. To combat this, we introduce a dynamic mechanism designed to make such ambiguity computationally untenable for complex axioms.
  This mechanism creates specific evidence "on-demand" when guarded rules are triggered. Its dynamic nature is crucial: the set of witnesses grows as the model learns. Consider the sample axiom $\forall x(\text{Pneumonia}(x) \rightarrow \exists y(\text{hasFinding}(x, y) \wedge \text{Fever}(y)))$.
  - **Early Training:** At epoch 1, model predictions are poor. The premise $[\![\text{Pneumonia}(c)]\!]$ is low for all constants, so no witnesses are created.
  - **Triggering Event:** As training progresses (e.g., epoch 50), the model learns to recognize specific entities, such as 'pneu$_{01}$', as instances of Pneumonia with high confidence:
  $$[\![\text{Pneumonia}(\text{pneu}_{01})]\!] \approx 0.95.$$
  - **Dynamic Creation:** This high-confidence guard triggers the creation of a *fresh* witness constant, $w_{\text{pneu,fever}}$, which is dynamically added to the domain.

  Each such event immediately augments the learning objective with a strong enforcement loss specific to that instance:
  $$\begin{aligned}&\mathcal{L}_{\text{witness}}(\text{axiom}, \text{pneu}_{01}) = \\ &1 - [\![\text{hasFinding}(\text{pneu}_{01}, w_{\text{pneu,fever}}) \wedge \text{Fever}(w_{\text{pneu,fever}})]\!]\end{aligned} \tag{6}$$
  Critically, this mechanism renders the **collapse to ambiguity** a prohibitively costly strategy. Since our framework interprets conjunction ($\wedge$) with the Product t-norm, an ambiguous output where both atoms in the conclusion have a truth value of 0.5 would yield a conjunction value of 0.25. With a high-confidence guard ($[\![\text{Pneumonia}(\text{pneu}_{01})]\!] \approx 1$), the witness loss becomes approximately $1 - 0.25 = 0.75$, a substantial penalty that the optimizer must eliminate. To minimize this loss, the model is mathematically compelled to push the truth values of *both* atoms towards 1, enforcing high-confidence predictions. These dynamically generated witnesses are fully integrated into the learning process. Their vector embeddings are initialized randomly (via Xavier) and become trainable parameters. Once added to Δ_{core}, they are available for sampling across all axioms, ensuring their semantic properties are constrained by the entire KB.

The entire **Core Domain** is thus the dynamically evolving set $\Delta_{\text{core}} = \Delta_H \cup \Delta_A \cup \Delta_G$.

4.3.2 Latent Domain Δ_{latent}: Driving Generalization. While the Core Domain Δ_{core} ensures logical fidelity by grounding the model in specific entities, relying on it alone risks overfitting. The model might memorize the properties of these particular constants without capturing the universal nature of the logical rules. To address this and ensure robust generalization, we introduce the **Latent Domain** Δ_{latent}, which represents the infinite, continuous embedding space itself, rather than a fixed set of discrete points.

Our strategy for utilizing the latent domain involves a continuous sampling mechanism. In each training step, we dynamically generate a mini-batch of "latent constants" by sampling vectors from a standard prior distribution (e.g., $\mathcal{N}(0, I)$ in $\mathbb{R}^d$). These temporary vectors act as random probes of the learned geometric space, used exclusively to evaluate universal axioms. This process defines our **generalization loss**, which forces the predicate networks to learn decision boundaries that are logically coherent everywhere in the embedding space, not just at the locations of known constants in Δ_{core}. Effectively, this ensures the model learns the general concept of what "a cat is", rather than merely learning that specific entities like "Tom is a cat".

4.4 The Hybrid Training Objective

The central learning objective in GuardNet is to find the optimal parameters θ that maximize the satisfiability of the KB $\mathcal{K}$. We derive our training objective directly from our principled fuzzy semantics defined in §3, constructing a loss function that minimizes the dissatisfaction of each axiom using our selected differentiable operators.

Loss for a Single Axiom Instance. For any universally quantified axiom of the form $\forall \bar{x}(\alpha \rightarrow \psi)$, the loss for a grounded instance is derived using the **Sigmoidal Reichenbach implication** recommended in §3.1. Let $I_R = 1 - [\![\alpha]\!] + [\![\alpha]\!][\![\psi]\!]$ be the standard Reichenbach implication score. The instance loss is defined as:

$$\begin{aligned}\mathcal{L}_{\text{instance}}(\forall \bar{x}(\alpha \rightarrow \psi); \bar{a}) &= 1 - [\![\alpha \rightarrow \psi]\!]_{\rho[\bar{x} \mapsto \bar{a}]} \\ &= 1 - \text{scale}\left(\sigma\left(s \cdot (I_R(\bar{a}) + b_0)\right)\right)\end{aligned} \tag{7}$$

where $\sigma(\cdot)$ is the Sigmoid function with steepness s and bias b_0. While this formulation introduces non-linearity to sharpen gradients around the decision boundary, it fundamentally preserves the deductive structure of the underlying Reichenbach logic (I_R). The core term I_R dictates that the penalty is driven by $[\![\alpha]\!](1 - [\![\psi]\!])$, aligning with the classical inference rules of Modus Ponens and Modus Tollens. By wrapping this logic in the Sigmoidal transformation, we ensure robust, non-vanishing gradients even for satisfied or violated constraints, as analyzed in App. A.3 (Tab. 5).

Theoretical Total Loss. The total loss aggregates the dissatisfaction over all axioms. Consistent with our choice of the **Generalized Mean** for quantifier aggregation (§3.2), we define the theoretical objective not as a simple linear expectation, but using the p-mean error aggregator $\mathcal{A}_p^{\forall}$:

$$\mathcal{L}_{\text{theoretical}}(\theta) = \sum_{\phi_i \in \mathcal{K}} \mathcal{A}_p^{\forall}\left(\left\{\mathcal{L}_{\text{instance}}(\phi_i; \bar{a}) \mid \bar{a} \in \Delta^{|\text{FV}(\phi_i)|}\right\}\right) \tag{8}$$

This formulation ensures that the model focuses on the most violated instances (behaving like a soft-max) rather than being diluted by the majority of satisfied cases, a critical feature for learning from sparse signals in KBs.

The Practical Hybrid Loss Function. To align with our **Hybrid Domain Strategy**, we realize this objective via stratified sampling. The total loss is a weighted combination:

$$\mathcal{L}_{\text{total}}(\theta) = \lambda \cdot \mathcal{L}_{\text{fidelity}} + (1-\lambda) \cdot \mathcal{L}_{\text{gen}} + \gamma \cdot \mathcal{L}_{\text{conf}} \tag{9}$$

where $\lambda \in (0, 1]$ balances fidelity and generalization, and γ controls the confidence regularization $\mathcal{L}_{\text{conf}} = \sum_i p_i(1-p_i)$. The two main loss terms are explicitly defined as follows:

- **Fidelity Loss ($\mathcal{L}_{\text{fidelity}}$):** Computed on the Core Domain Δ_{core}, this term enforces consistency with known facts and dynamic witnesses:

$$\mathcal{L}_{\text{fidelity}} = \mathcal{A}_p^{\forall}\left(\{\mathcal{L}_{\text{instance}}(\phi; \bar{c}) \mid \bar{c} \sim \Delta_{\text{core}}\}\right) + \sum \mathcal{L}_{\text{witness}} \tag{10}$$

  It acts as "textbook examples", grounding the model in the given theory and explicitly preventing collapse via the witness term.
- **Generalization Loss ($\mathcal{L}_{\text{gen}}$):** Computed on the Latent Domain Δ_{latent}, this term enforces universal consistency:

$$\mathcal{L}_{\text{gen}} = \mathcal{A}_p^{\forall}\left(\{\mathcal{L}_{\text{instance}}(\phi; \bar{z}) \mid \bar{z} \sim \mathcal{N}(0, I)\}\right) \tag{11}$$

  It acts as the "final exam", testing the model with countless unseen inputs to ensure it has learned the universal principles rather than memorizing specific constants.

By jointly optimizing these complementary loss terms, GuardNet learns a model that is both faithful to the provided data and robustly generalizable.

Theorem 1 (Zero-Loss Semantic Consistency). *Let $\mathcal{K}$ be a KB consisting of a finite set of GF formulas. If a GuardNet model trained on $\mathcal{K}$ achieves $\mathcal{L}_{\text{fidelity}}(\theta) = 0$, then the learned fuzzy interpretation $\mathcal{G}_\theta$ satisfies all axioms on the Core Domain Δ_{core}. That is, for every axiom $\phi \in \mathcal{K}$ and every grounding $\bar{c} \in \Delta_{\text{core}}^{|\text{FV}(\phi)|}$, we have $[\![\phi]\!]_{\mathcal{G}_\theta, \bar{c}} = 1$.*

Thm. 1 establishes that the loss function and the logical semantics are formally aligned: zero loss implies full satisfaction. This is a consistency guarantee for the training objective, not a claim about the practically relevant nonzero-loss regime. The following corollary addresses the nonzero-loss setting, while the main scalability guarantee is provided by the complexity result in Thm. 2.

Corollary 1 (Approximate satisfaction). *For axiom $\phi_i = \forall \bar{x}\,(\alpha_i \rightarrow \psi_i)$, define the instance dissatisfaction $z_{i,c} = 1 - [\![\alpha_i \rightarrow \psi_i]\!]_c$ for $c \in \text{ext}_{\mathcal{K}}(\alpha_i)$, and let $k_i = |\text{ext}_{\mathcal{K}}(\alpha_i)| > 0$. If $\mathcal{E}_p^{\forall}(\mathbf{z}_i) \leq \varepsilon_i$ for $p \geq 1$, then:*

$$\text{(i)}\ \frac{1}{k_i} \sum_{c \in \text{ext}_{\mathcal{K}}(\alpha_i)} z_{i,c} \leq \varepsilon_i \qquad \text{(ii)}\ \max_{c \in \text{ext}_{\mathcal{K}}(\alpha_i)} z_{i,c} \leq \min\{1,\ k_i^{1/p}\,\varepsilon_i\}$$

Axioms with $k_i = 0$ are treated as vacuously satisfied under the masked grounding semantics and contribute zero universal grounding loss. Note that the global fidelity loss $\frac{1}{m}\sum_i \varepsilon_i \leq \varepsilon$ controls the average over axioms, not each ε_i uniformly; uniform per-axiom guarantees require controlling each ε_i separately.

Theorem 2 (Training Complexity). *Let $\mathcal{K}$ be a KB with m guarded axioms $\{\forall \bar{x}_i(\alpha_i \rightarrow \psi_i)\}_{i=1}^m$ over a domain of N entities. Let d denote the embedding dimension, L the number of layers in predicate MLPs, B the batch size, and $|\mathcal{P}|$ the number of predicates. Recall that $\text{ext}(\alpha)$ denotes the guard extension (defined in §4.1) and Δ_{core} the Core Domain (§4.3). A single training iteration of GuardNet has:*

Time Complexity:

$$O\left(\sum_{i=1}^{m} |\text{ext}(\alpha_i)| \cdot L \cdot d^2\right) = O\left(m \cdot |E|_{\max} \cdot L \cdot d^2\right) \tag{12}$$

where $|E|_{\max} = \max_i |\text{ext}(\alpha_i)|$ is the largest guard extension. The d^2 factor arises from matrix operations in the MLP forward pass.

Space Complexity:

$$O\left(|\Delta_{\text{core}}| \cdot d + |\mathcal{P}| \cdot L \cdot d^2 + B \cdot |E|_{\max}\right) \tag{13}$$

where the three terms account for entity embeddings, predicate MLP parameters, and the batch of grounded instances, respectively.

*In sparse KBs where $|E|_{\max} = O(N)$, both complexities are **linear in the domain size** N, circumventing the $O(N^2)$ bottleneck of unguarded approaches.*

5 Empirical Evaluation

In the previous sections, we theoretically argued that GuardNet resolves the fundamental conflict between expressivity and scalability. This is achieved through our novel **Guarded Inductive Bias**, which restricts computation to valid topological neighborhoods, and our **Hybrid Domain Strategy**, which ensures robust generalization. This section provides the **empirical substantiation** of these claims. We structure our evaluation to progress from system-level feasibility to task-level performance, and finally to architectural component analysis. Specifically, we design our experiments to answer three core Research Questions (RQs):

- **RQ1 (Scalability & System Efficiency):** Can GuardNet physically overcome the "tensor broadcasting" bottleneck analyzed in §4.1 and scale to large KBs (e.g., SNOMED CT) where existing neural-symbolic baselines fail?
- **RQ2 (Comparative Performance):** Once feasibility is established, does GuardNet outperform state-of-the-art Geometric Embeddings, GNNs, and NeSy models across diverse reasoning tasks, demonstrating robust generalization capabilities?
- **RQ3 (Ablation & Mechanism Analysis):** What are the sources of these performance gains? We conduct a comprehensive ablation study to verify the impact of the **Hybrid Domain Strategy** in preventing representational collapse, and analyze the sensitivity of learning dynamics to different **Fuzzy Operators**.

Data Translation Strategy. Since no native benchmark datasets exist specifically for the GF, we construct our experimental testbed by converting standard ontologies into GF formulas. This approach is theoretically well-grounded due to the deep syntactic connections among these formalisms. GF was originally introduced in [3] as a natural generalization of modal logic to the polyadic (multi-variable) setting, preserving key properties such as decidability and the finite model property. DLs, in turn, are well-known to be notational variants of modal logics [6]. As a result, standard DL axioms translate directly into GF sentences. This principled translation ensures that our experiments faithfully evaluate reasoning capabilities on semantically equivalent logical structures.

Evaluation Tasks and Datasets. We evaluate GuardNet on two reasoning tasks. **TBox Completion** tests concept subsumption:

given a test axiom $C \sqsubseteq D$, we rank the true axiom against corruptions generated by replacing C (or D) with all candidate concepts. **ABox Completion** tests link prediction [45]: given a test triple $r(h, t)$, we rank candidates for head prediction $(?, r, t)$ and tail prediction $(h, r, ?)$. Both tasks employ the standard **filtered setting** [11], removing known true facts from the candidate list.

For TBox completion, we use **SNOMED CT** (377K concepts) [48] as a scalability benchmark in medical terminology and **Gene Ontology (GO)** (44K concepts) [4] for its hierarchically rich biological taxonomy. These ontologies are expressed in the DL $\mathcal{EL}^{++}$ [5], whose core axiom patterns (concept inclusion, existential restriction, conjunction, role hierarchy) translate directly into GF sentences (see App. C). For ABox completion, we use two protein-protein interaction (PPI) datasets (**Yeast PPI**: 6.4K entities, 110K edges, **Human PPI**: 17.6K entities, 75K edges), combining factual interactions from the STRING database [52] with TBox constraints from GO. Following standard conventions, all datasets are randomly split into 80% training, 10% validation, and 10% test sets.

GF Coverage and Non-Guarded Axioms. The only $\mathcal{EL}^{++}$ constructs that fall outside GF are role chains ($R_1 \circ R_2 \sqsubseteq R_3$) and transitivity declarations, both of which require three variables with no single binary guard atom. In practice, these constitute a negligible fraction of our benchmarks: SNOMED CT contains only 4 such axioms out of 391K total TBox axioms, and GO similarly few. All remaining axioms receive full $O(|\mathcal{E}|)$ guarded grounding. For the excluded axioms, GuardNet applies bounded-depth neighborhood sampling as a fallback, incurring the same $O(N^3)$ worst-case cost that any NeSy system pays for three-variable formulas. The overall complexity remains dominated by the guarded majority.

Baselines. We benchmark GuardNet against 15 baselines spanning five distinct paradigms in KBC tasks:

- **Geometric Ontology Embeddings:** EL Embeddings (ELEM) [27], Box^{2}EL [24], and TransBox [66]. As state-of-the-art approaches that are specifically designed to capture the semantics of the DL $\mathcal{EL}^{++}$, these constitute the strongest and most direct baselines for ontology-based reasoning tasks.
- **Graph Neural Networks (GNNs):** CompGCN [59], GRAIL [55], and NBFNet [73]. These represent the leading GNN-based approaches for the KBC task, whose reliance on implicit message propagation provides a crucial contrast to GuardNet's explicit, logic-defined neighborhoods.
- **Neural-Symbolic (FOL-based):** LTN [7], logLTN [8], and Neural LP [65]. Rooted in more general, unguarded FOL, these frameworks highlight the computational complexity that GuardNet's principled restriction to GF is designed to overcome.[1]
- **Probabilistic Frameworks:** NeurASP [67], DeepProbLog [34], and DeepStochLog [62]. These are prominent probabilistic logic programming systems that integrate neural networks with probabilistic inference, representing an orthogonal approach to neural-symbolic integration.

[1]Note that to enable these baselines to run on large datasets for Tab. 2, we applied stochastic mini-batching. As discussed in §5.1, this engineering compromise avoids OOM errors but severely degrades logical reasoning performance.

- **Standard KG Embedding Models:** TransE [11], RotatE [51], and ComplEx [56]. These widely adopted embedding methods learn logical patterns *implicitly* through geometric formulations, providing a clear baseline to quantify the performance gains from explicit logical reasoning.

Evaluation Metrics. For both completion tasks, we report **Hits@K ($K \in \{1, 10, 100\}$)** and **MRR**. GuardNet ranks candidates by fuzzy satisfaction degree $[\![\alpha]\!] \in [0, 1]$ in descending order; geometric ontology embedding baselines rank by ascending distance, and KG embedding models use their respective scoring functions.

Implementation Details. All experiments were run on NVIDIA RTX 4090 GPUs using PyTorch 2.0 with CUDA 11.8. For all models, the AdamW optimizer [33] was used with an early stopping strategy (patience of 15 epochs) based on validation MRR.

For training, we employ **negative sampling** with $\omega = 64$ corrupted samples per positive instance. Within our fuzzy semantics, observed facts $P(a, b)$ are assigned target truth value 1, while corrupted facts $P(a', b)$ or $P(a, b')$ are assigned target truth value 0. This discriminative signal complements the axiom-based losses in Equation 9: while $\mathcal{L}_{\text{fidelity}}$ and $\mathcal{L}_{\text{gen}}$ enforce logical consistency across the embedding space, the negative sampling loss provides explicit supervision for individual facts. We adopt a self-adversarial margin-based loss (margin $\delta = 1.0$) to sharpen the discrimination boundary. L2 regularization ($\lambda_{\text{reg}} = 10^{-5}$) is applied to all embeddings.

All hyperparameters were selected via grid search with sensitivity analysis on validation sets. For baselines, search spaces included learning rates $\in \{10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}\}$, embedding dimensions $d \in \{100, 200, 400\}$, and batch sizes $\in \{256, 512, 1024\}$. For GuardNet, the optimal configuration is: learning rate 2×10^{-4}, embedding dimension $d = 200$, batch size 512 (SNOMED CT) or 1024 (others), predicate MLPs with two 256-unit ReLU layers. For fuzzy operators, we use steepness $s = 10$ and bias $b_0 = -0.5$ in the Sigmoidal implication, and aggregation exponent $p = 2$ for quantifiers. The hybrid loss weights are $\lambda = 0.7$ (fidelity-generalization balance) and $\gamma = 0.01$ (confidence penalty). These values were determined through systematic sensitivity analysis.

To guarantee reproducibility, all reported metrics are the **mean ± standard deviation over 5 independent runs** with different random seeds. Paired t-tests between GuardNet and the strongest baseline confirm statistical significance ($p < 0.01$) across all benchmarks and metrics, except for H@1 on SNOMED CT ($p < 0.05$).

5.1 Scalability and System Efficiency (RQ1)

We begin by addressing the most critical bottleneck in NeSy reasoning: physical scalability. To answer **RQ1**, we conducted a stress test comparing GuardNet against representative NeSy baselines (LTN [7], logLTN [8], and Neural LP [65]) on increasingly large subsets of SNOMED CT.

The results in **Fig. 5** offer definitive validation of our polynomial complexity advantages. Crucially, we must distinguish between computational feasibility and logical validity. While baselines can technically run on smaller-scale ABox tasks (as reported in Tab. 2) by employing aggressive stochastic mini-batching, this strategy fundamentally breaks the global dependency chains required for

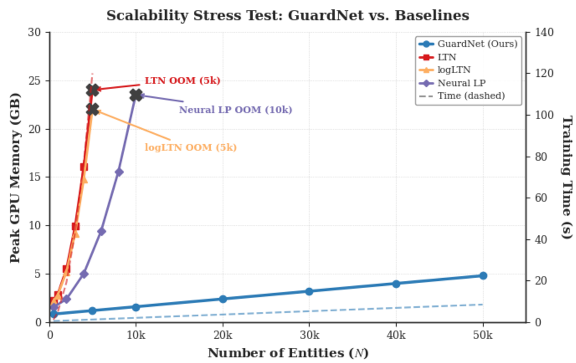

Figure 5: We compare the peak GPU memory usage (solid lines) and training time (dashed) as the number of entities N increases. The "OOM Wall": Standard NeSy models LTN (red) and logLTN (orange) hit the memory limit (24GB) at just 5K entities due to their $O(N^2)$ tensor broadcasting bottleneck. Even matrix-based approaches like Neural LP (purple) fail at 10K. The Guarded Advantage: In contrast, GuardNet (blue) maintains a linear memory footprint $O(|\mathcal{E}|)$ via topological indexing, scaling effortlessly to 50K+ entities.

logical inference. As shown in Tab. 2, even with such engineering compromises, NeSy baselines completely fail (DNF) on large-scale TBox reasoning tasks such as SNOMED CT and GO, while achieving only moderate performance on the smaller PPI datasets (e.g., LTN attains MRR of 0.285 on Yeast PPI, far below GuardNet's 0.405). **Fig. 5** reveals the root cause: when forced to perform *rigorous* full grounding to preserve logic, these models hit a quadratic memory wall at just 5K entities. In sharp contrast, GuardNet maintains linear scalability without sacrificing logical fidelity, achieving the unique combination of SOTA accuracy and linear efficiency.

5.2 Comparative Performance (RQ2)

Having established physical feasibility, we evaluate the predictive performance of GuardNet on full-scale datasets. Tab. 2 presents the "Grand Slam" benchmark results, comparing our model against 15 state-of-the-art baselines across TBox (Concept Subsumption) and ABox (Link Prediction) tasks.

GuardNet demonstrates consistent superiority across all datasets and reasoning tasks. On the massive **SNOMED CT** ontology, GuardNet attains an MRR of 0.125, surpassing the strongest geometric baseline TransBox (0.119) and Box^{2}EL (0.114), while the leading GNN baseline NBFNet lags significantly behind (0.055) due to its inability to capture strict logical entailment in deep hierarchies. This advantage extends to the **GO**, where GuardNet achieves an MRR of 0.133, outperforming TransBox (0.126) by a notable margin, while NeSy baselines like LTN fail to complete (DNF) on this scale.

The primary reason for this success lies in our framework's **high logical fidelity**. Unlike geometric models that approximate logical constraints through spatial relationships (e.g., box containment or vector rotation), GuardNet's loss function is derived directly from the **GF semantics**. This forces the learned embeddings to maintain strict consistency with the ontological structure, effectively pruning the search space to valid neighborhoods. This is particularly evident in precision-oriented metrics, where GuardNet excels by avoiding the false positives that plague purely statistical methods.

On ABox reasoning tasks (Yeast and Human PPI), the dramatic performance differential with standard KG embedding models confirms the fundamental importance of incorporating structured logical reasoning. GuardNet achieves an MRR of 0.405 on Yeast PPI, significantly outperforming the leading GNN baseline NBFNet (0.351) by over 15%. This confirms that the *Guarded Inductive Bias* is not limited to taxonomic reasoning but also generalizes robustly to inductive link prediction, effectively capturing multi-hop logical dependencies that standard message-passing schemes miss.

5.3 Ablation and Mechanism Analysis (RQ3)

To isolate the source of the performance gains, we perform a comprehensive component analysis and investigate the model's sensitivity to operator choices.

Hybrid Domain Strategy. Tab. 3 reveals that the **Hybrid Domain Strategy** is critical for preventing representational collapse.

- **Preventing Emptiness:** The base model (Row 2) relying solely on the Herbrand domain (Δ_H) suffers from an 82% collapse rate on SNOMED CT, as the model learns to predict logical falsity to vacuously satisfy axioms. Introducing **Atom Witnesses** (Δ_A, Row 3) provides the necessary static anchors, drastically reducing collapse to 12% and nearly doubling the MRR.
- **Enforcing Decisiveness:** The addition of **Guarded Witnesses** (Δ_G, Row 4) further refines the decision boundaries, handling dynamic triggering of rules. Finally, the full model with **Latent Generalization** (Δ_{latent}, Row 5) eliminates collapse almost entirely (<1%), proving that continuous sampling is essential for learning universal logical principles.

- **Role of Learnable Embeddings:** Row 6 isolates the contribution of joint embedding optimization by freezing TransE-pretrained embeddings. The Hybrid Domain Strategy remains fully active, so collapse rates stay below 1%, confirming that collapse prevention is independent of embedding trainability. However, MRR drops by ≈20% on TBox tasks (SNOMED CT, GO), where pretrained relational embeddings lack alignment with concept-hierarchy axioms and the frozen geometry cannot adapt. On PPI tasks, where the pretrained structure already encodes relational patterns, the drop is only ≈8%. This asymmetry demonstrates that logic-driven embedding fine-tuning is essential for concept-level reasoning but less critical for relation-level link prediction.

Mechanism Analysis: Visualizing Collapse. To visually confirm the elimination of representational collapse, we plotted the density distribution of predicted truth values on the test set in **Fig. 6**.

The **Unguarded** baseline (red curve) exhibits a pathological distribution heavily skewed towards 0. This empirically validates our theoretical concern that without proper grounding, the model minimizes loss by learning "vacuous truths" (i.e., predicting all atoms as False to trivially satisfy implications like False $\rightarrow \cdots$). In contrast, GuardNet (blue curve) displays a healthy **bimodal distribution** with distinct peaks near 0 (False) and 1 (True). This indicates that our Hybrid Domain Strategy successfully forces the model to make

Table 2: Overall KBC Performance Comparison. We evaluate GuardNet against SOTA baselines across four benchmarks spanning TBox reasoning (concept subsumption) and ABox reasoning (link prediction). GuardNet (Product t-norm + Sigmoidal Reichenbach implication) achieves the best performance across all metrics. DNF: Did Not Finish (OOM/Timeout after 72h).

Model / Configuration	SNOMED CT (377K) TBox Reasoning (Concept Subsumption)				GO (44K) TBox Reasoning (Concept Subsumption)				Yeast PPI (110K edges) ABox Reasoning (Link Prediction)				Human PPI (75K edges) ABox Reasoning (Link Prediction)			
	H@1	H@10	H@100	MRR	**H@1**	H@10	H@100	MRR	**H@1**	H@10	H@100	MRR	**H@1**	H@10	H@100	MRR
GuardNet (Prod+Sig)	**5.1±.1**	**28.3±.4**	**70.5±.3**	**.125±.00**	**5.5±.2**	**29.8±.3**	**73.4±.2**	**.133±.00**	**30.5±.4**	**60.2±.5**	**91.1±.3**	**.405±.00**	**29.2±.5**	**57.9±.6**	**88.9±.4**	**.388±.00**
Geometric Ontology Embedding Models																
ELEM	2.1±.1	20.1±.6	38.9±.5	.078±.00	2.4±.1	23.8±.5	43.4±.4	.089±.00	20.5±.6	45.0±.8	74.8±.6	.301±.00	18.2±.8	40.1±1.1	69.7±.8	.268±.00
Box²EL	3.4±.2	25.5±.6	68.1±.5	.114±.00	3.9±.2	26.5±.5	70.9±.4	.120±.00	27.1±.5	55.1±.6	86.9±.5	.368±.00	25.5±.6	52.3±.8	82.7±.6	.346±.00
TransBox	3.6±.2	26.8±.5	69.2±.4	.119±.00	4.1±.2	27.8±.4	71.8±.3	.126±.00	28.5±.5	57.5±.6	88.5±.4	.385±.00	26.8±.6	54.8±.7	85.2±.5	.365±.00
Graph Neural Networks																
CompGCN	0.8±.1	8.1±.5	20.3±.4	.041±.00	1.2±.1	9.9±.4	24.1±.3	.052±.00	23.5±.6	50.5±.7	78.3±.5	.328±.00	21.5±.7	45.2±.8	72.1±.6	.302±.00
GRAIL	1.0±.1	9.8±.5	24.0±.5	.050±.00	1.5±.1	11.9±.5	28.1±.4	.063±.00	25.8±.6	52.6±.7	80.2±.5	.345±.00	23.8±.7	48.0±.8	75.5±.6	.321±.00
NBFNet	1.2±.1	10.5±.4	25.8±.4	.055±.00	1.8±.1	13.0±.4	30.2±.3	.071±.00	26.5±.5	53.5±.6	81.1±.5	.351±.00	24.5±.6	49.1±.7	76.9±.6	.332±.00
Expressive NeSy Models (Unguarded, FOL-based)																
LTN (Prod+Sig)	**DNF**				**DNF**				17.5±.5	42.5±.9	70.1±.7	.285±.00	16.0±.6	39.8±1.1	65.4±.8	.265±.01
logLTN (Prod+Sig)	**DNF**				**DNF**				19.5±.6	45.1±1.1	72.7±.9	.302±.01	18.5±.7	42.0±1.3	68.2±.9	.284±.01
Neural LP	**DNF**				**DNF**				18.0±.5	43.3±.9	71.1±.8	.291±.00	16.5±.6	40.5±1.2	66.8±.9	.272±.01
Probabilistic Frameworks																
NeurASP	**DNF**				**DNF**				14.1±.3	35.5±.6	60.4±.5	.225±.00	12.8±.4	32.8±.7	58.5±.6	.208±.00
DeepProbLog	**DNF**				**DNF**				12.8±.2	32.2±.5	58.1±.4	.208±.00	11.5±.3	29.5±.6	55.8±.5	.192±.00
DeepStochLog	**DNF**				**DNF**				16.5±.4	38.8±.7	65.2±.6	.255±.00	15.1±.5	36.2±.8	62.9±.7	.235±.00
Standard KG Embedding Models																
TransE	0.2±.1	2.1±.3	8.7±.2	.018±.00	0.3±.1	2.8±.3	12.3±.2	.023±.00	15.5±.4	38.0±.7	64.9±.5	.238±.00	14.0±.5	35.0±.8	60.7±.6	.219±.00
RotatE	0.3±.1	3.2±.4	11.8±.3	.025±.00	0.4±.1	4.1±.4	16.7±.3	.032±.00	18.5±.5	42.6±.8	68.1±.6	.268±.00	16.0±.6	38.3±.9	64.3±.7	.245±.00
ComplEx	0.3±.1	2.6±.3	10.1±.3	.022±.00	0.4±.1	3.5±.3	14.9±.3	.028±.00	17.0±.5	40.5±.8	66.4±.6	.252±.00	15.5±.6	36.1±.9	62.8±.7	.232±.00

Table 3: Comprehensive Component Analysis (RQ3). We systematically dissect GuardNet's performance gains. Col. (Collapse Rate): % of predictions stuck in trivial states. Low is good. RSR (Rule Satisfaction Rate): Average truth value of axioms. High is good. FPE (Row 6): Fixed Pretrained Embedding (TransE-initialized, frozen during training). Note: On PPI (Graph tasks), Collapse Rate is consistently low (<1%) for all GuardNet variants due to explicit positive edges, but RSR steadily improves, showing that the model learns to satisfy logical rules more strictly. Percentage improvements are computed relative to Row 2 (GuardNet Base) for TBox tasks and Row 1 (LTN Optimized) for ABox tasks.

Model Configuration	Domain	SNOMED CT						GO						Yeast PPI					Human PPI				
		H@1	H@10	H@100	MRR	**Col.**	**RSR**	**H@1**	H@10	H@100	MRR	**Col.**	**RSR**	**H@1**	H@10	H@100	MRR	**RSR**	**H@1**	H@10	H@100	MRR	**RSR**
1. LTN (Optimized)	**Unguard.**	**DNF (OOM)**						**DNF (OOM)**						17.5	42.5	70.1	.285	.82	16.0	39.8	65.4	.265	.80
(Product + Sigmoidal)	(Δ_H)																						
2. GuardNet (Base)	**Guarded**	1.2	11.5	24.8	.055	82%	.65	1.4	12.8	26.5	.062	79%	.68	22.1	49.8	80.5	.335	.91	20.5	46.5	75.2	.312	.89
(Product + Sigmoidal)	(Δ_H)	*(High Collapse due to Emptiness)*						*(High Collapse due to Emptiness)*						+26%	+17%	+14%	+17%	+11%	+28%	+16%	+15%	+17%	+11%
3. + Atom Witness	**Hybrid**	3.8	23.5	61.5	.105	**12%**	.92	4.2	25.8	65.4	.115	**10%**	.93	27.5	55.2	86.8	.375	.96	26.2	53.5	82.5	.352	.94
(+ Static Anchors)	($\cup\Delta_A$)	+216%	+104%	+148%	+91%	↓70pt	+.27	+200%	+101%	+146%	+85%	↓69pt	+.25	+24%	+10%	+7.8%	+12%	+.05	+27%	+15%	+9.7%	+13%	+.05
4. + Guarded Witness	**Hybrid**	4.8	26.8	68.2	.118	5%	.96	5.1	28.2	71.5	.126	4%	.97	29.8	59.1	90.5	.396	.98	28.5	57.0	87.5	.378	.96
(+ Dynamic Anchors)	($\cup\Delta_G$)	+26%	+14%	+11%	+12%	↓7pt	+.04	+21%	+9.3%	+9.3%	+9.6%	↓6pt	+.04	+8.3%	+7.1%	+4.2%	+5.6%	+.02	+8.7%	+6.5%	+6.0%	+7.3%	+.02
5. Full GuardNet	**Hybrid**	**5.1**	**28.3**	**70.5**	**.125**	**<1%**	**.99**	**5.5**	**29.8**	**73.4**	**.133**	**<1%**	**.99**	**30.5**	**60.2**	**91.1**	**.405**	**.99**	**29.2**	**57.9**	**88.9**	**.388**	**.98**
(+ Latent Generalization)	($\cup\Delta_{\text{latent}}$)	+6.2%	+5.6%	+3.4%	+5.9%	↓4pt	+.03	+7.8%	+5.7%	+2.7%	+5.6%	↓3pt	+.02	+2.3%	+1.9%	+0.7%	+2.3%	+.01	+2.4%	+1.6%	+1.6%	+2.6%	+.02
6. FPE	**Hybrid**	4.0	22.5	57.8	.100	<1%	.95	4.4	24.2	60.8	.109	<1%	.95	27.8	55.8	87.2	.372	.97	26.8	53.8	84.5	.357	.96
(TransE, frozen)	(Δ_{core} $\cup\,\Delta_{\text{latent}}$)	−22%	−20%	−18%	−20%	·	−.04	−20%	−19%	−17%	−18%	·	−.04	−8.9%	−7.3%	−4.3%	−8.1%	−.02	−8.2%	−7.1%	−5.0%	−8.0%	−.02

decisive, semantically meaningful predictions, effectively avoiding the collapse modes that plague standard NeSy approaches.

Sensitivity to Fuzzy Operators. Finally, **Fig. 7** highlights the pivotal role of the **Sigmoidal Implication**. While the choice of T-norm (rows) shows marginal variance, the implication operator (columns) is decisive. Standard R-implications (Goguen) frequently lead to vanishing gradients ("Dead Zones"), causing models to fail (DNF) or underperform on large datasets. The Sigmoidal Implication (Red Box) consistently resolves this, providing the robust gradient signal required for deep reasoning architectures to converge.

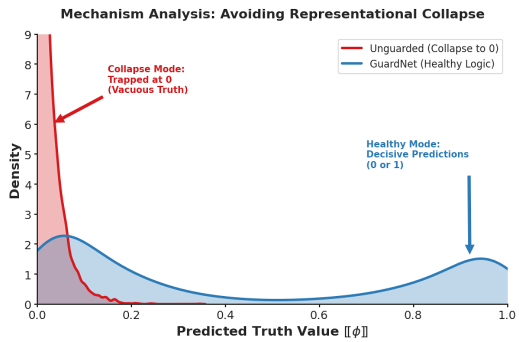

Figure 6: Unguarded model (red line) collapses to vacuous truths (trapped at 0), whereas GuardNet (blue line) maintains a healthy bimodal distribution with decisive predictions (near 0 or 1), effectively avoiding the collapse modes.

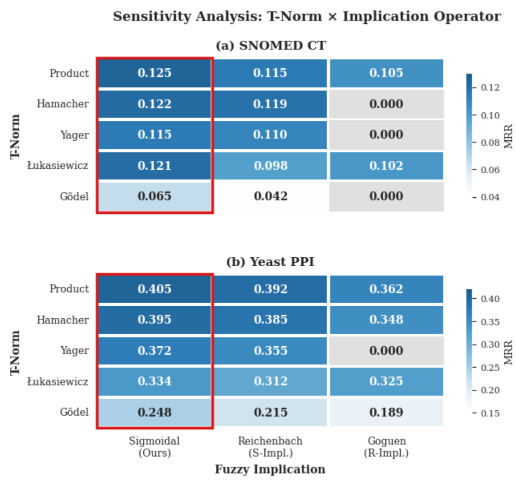

Figure 7: The heatmap isolates the impact of fuzzy operators on MRR. While the choice of T-norm (rows) has marginal variance, the choice of Implication (columns) is decisive.

Sensitivity of Sigmoidal Parameters. Fig. 8 analyzes the sensitivity to steepness s and bias b_0 of the Sigmoidal Reichenbach implication. Panel (a) shows that performance peaks at (s=10, b_0=−0.5) with a broad plateau across $s \in [5, 20]$ and $b_0 \in [-0.75, -0.25]$, confirming robustness. Panel (b) reveals the interplay between s and training dynamics at fixed b_0=−0.5: as s increases from 1 to 10, MRR improves and collapse rate drops below 1%, while $|\Delta_G|$ decreases from ≈22.8k to ≈12.4k as sharper gradients enable more precise witness creation. Beyond s=20, the sigmoid approaches a step function and gradients vanish away from the boundary, degrading both MRR and collapse rate.

6 Conclusion and Future Work

In this work, we resolved the longstanding dichotomy between logical expressivity and computational scalability in neural-symbolic

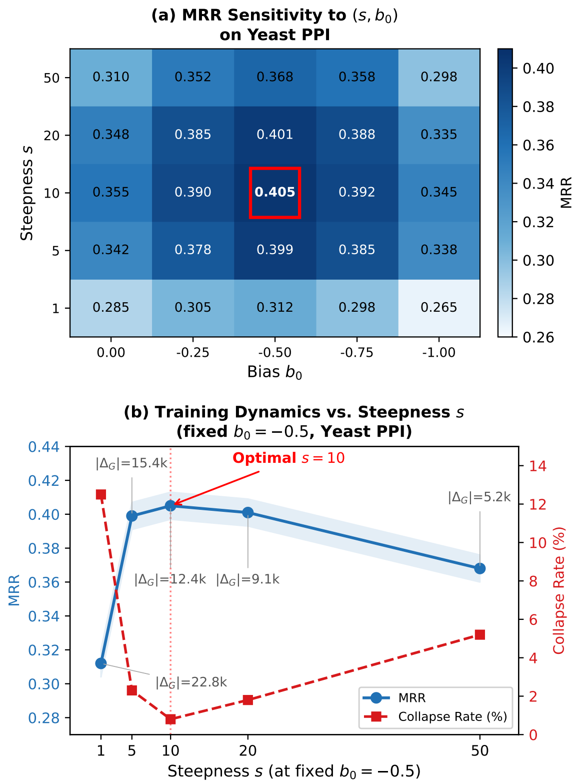

Figure 8: Sensitivity of Sigmoidal Implication Parameters on Yeast PPI. (a) MRR heatmap across (s, b_0) grid; red box marks the optimal (s=10, b_0=−0.5). (b) MRR and Collapse Rate vs. steepness s at fixed b_0=−0.5, with $|\Delta_G|$ annotations.

learning. By reinterpreting the **Guarded Fragment** of full FOL not as a restriction, but as a powerful topological inductive bias, we fundamentally restructured the computational graph of reasoning: transforming intractable global search into efficient, neighborhood-constrained lookups. This approach neuralizes the tractability mechanism of GTGDs from database theory, establishing a formal bridge between classical query answering and differentiable neural reasoning. GuardNet demonstrates that rigorous logical constraints need not be computational bottlenecks; on the contrary, the "guard" acts as a semantic index that guides neural optimization, enabling linear scalability ($O(|\mathcal{E}|)$) on massive ontologies like SNOMED CT where prior methods fail. Furthermore, our Hybrid Domain Strategy successfully reconciles data fidelity with generalization, effectively preventing representational collapse.

For future work, we envision two strategic directions:

- **In-Database Reasoning:** Integrating GuardNet's sparse grounding kernels directly into graph database engines (e.g., logic-aware stored procedures) to support real-time, zero-shot logical query answering without external model export.
- **Beyond Static Graphs:** Extending the guarded mechanism to *Temporal Guarded Logic* to handle dynamic knowledge graphs where validity intervals act as temporal guards.

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A Theoretical Proofs and Analysis

In this appendix, we provide the formal definitions, rigorous mathematical proofs, and detailed complexity analysis for the theorems presented in the main text. We begin by summarizing the notation used throughout the paper.

A.1 Summary of Notation

Table 4: Summary of Mathematical Notation

Symbol	Description
$\mathcal{K} = (\mathcal{T}, \mathcal{A})$	Knowledge Base composed of TBox $\mathcal{T}$ and ABox $\mathcal{A}$
$\Sigma = (\mathcal{P}, C)$	Signature containing predicates $\mathcal{P}$ and constants C
$\Delta, \Delta_{\text{core}}$	The domain of interpretation and the Core Domain subset
$\text{ext}(\alpha)$	The extension of a guard atom α (set of satisfying tuples in $\mathcal{K}$)
$N, \lvert\mathcal{E}\rvert$	Number of entities ($\lvert\Delta\rvert$) and number of edges/facts in $\mathcal{K}$
d, L	Embedding dimension and number of MLP layers
$\mathbf{e}_c$	Vector embedding of constant $c \in \mathbb{R}^d$
$[\![\phi]\!] \in [0, 1]$	Fuzzy truth value of formula ϕ
$T_P(a, b)$	Product T-norm (Fuzzy Conjunction): $a \cdot b$
$I_{\sigma R}(a, b)$	Sigmoidal Reichenbach Implication
$\mathcal{A}_p^{\forall}$	Universal Quantifier (Truth) Aggregator (as defined in Main Text Equation 2)
$\mathcal{E}_p^{\forall}$	Error Aggregator (Loss counterpart used in proofs, $\mathcal{E} = 1 - \mathcal{A}$)

A.2 Formal Semantics of GF

The following definition, referenced from §2, provides the complete formal semantics of GF for readers requiring the full technical detail.

Definition 3 (Semantics of GF). *A **structure** for a signature Σ is a pair $\mathfrak{A} = \langle \Delta, \cdot^{\mathfrak{A}} \rangle$, where Δ is a non-empty set called the **domain**, and $\cdot^{\mathfrak{A}}$ is an **interpretation function** that maps each constant $c \in C$ to an element $c^{\mathfrak{A}} \in \Delta$ and each n-ary predicate $P \in \mathcal{P}$ to a relation $P^{\mathfrak{A}} \subseteq \Delta^n$. A **variable assignment** is a function $\rho : \mathcal{V} \to \Delta$. For a tuple of variables $\bar{x} = (x_1, \ldots, x_k)$ and elements $\bar{a} = (a_1, \ldots, a_k) \in \Delta^k$, we write $\rho[\bar{x} \mapsto \bar{a}]$ for the assignment that maps each x_i to a_i and agrees with ρ elsewhere. The **interpretation** of a term t under $\mathfrak{A}$ and ρ, denoted $t^{\mathfrak{A},\rho}$, is defined by: $x^{\mathfrak{A},\rho} = \rho(x)$ for variables, and $c^{\mathfrak{A},\rho} = c^{\mathfrak{A}}$ for constants.*

*The **satisfaction relation** $\mathfrak{A}, \rho \models \phi$ is defined inductively:*

- $\mathfrak{A}, \rho \models P(t_1, \ldots, t_n)$ *iff* $(t_1^{\mathfrak{A},\rho}, \ldots, t_n^{\mathfrak{A},\rho}) \in P^{\mathfrak{A}}$.
- $\mathfrak{A}, \rho \models \neg\phi$ *iff* $\mathfrak{A}, \rho \not\models \phi$.
- $\mathfrak{A}, \rho \models \phi \wedge \psi$ *iff* $\mathfrak{A}, \rho \models \phi$ *and* $\mathfrak{A}, \rho \models \psi$.
- $\mathfrak{A}, \rho \models \exists\bar{x}\,(\alpha \wedge \psi)$ *iff there exists* $\bar{a} \in \Delta^{|\bar{x}|}$ *such that* $\mathfrak{A}, \rho[\bar{x} \mapsto \bar{a}] \models \alpha \wedge \psi$.
- $\mathfrak{A}, \rho \models \forall\bar{x}\,(\alpha \to \psi)$ *iff for all* $\bar{a} \in \Delta^{|\bar{x}|}$, $\mathfrak{A}, \rho[\bar{x} \mapsto \bar{a}] \models \alpha$ *implies* $\mathfrak{A}, \rho[\bar{x} \mapsto \bar{a}] \models \psi$.

The remaining Boolean connectives ($\vee, \to$) are interpreted standardly.

A.3 Full Fuzzy Operator Comparison

Tab. 5 provides the comprehensive comparison of all candidate fuzzy operators referenced in §3.

A.4 Formal Definitions of Fuzzy Operators

To support the soundness proof, we explicitly define the properties of the differentiable fuzzy operators used in GuardNet. These definitions are strictly aligned with the formulations in §3.2 of the main text.

Definition 4 (Product T-norm). *The conjunction of two fuzzy truth values $a, b \in [0, 1]$ is defined as:*

$$T_P(a, b) = a \cdot b \tag{14}$$

Property: $T_P(a, b) = 1$ *if and only if* $a = 1$ *and* $b = 1$.

Definition 5 (Sigmoidal Reichenbach Implication). *The standard Reichenbach implication is $I_R(a, c) = 1 - a + a \cdot c$. The **Sigmoidal Reichenbach** implication used in Equation 7 is defined as:*

$$I_{\sigma R}(a, c) = \text{scale}\,(\sigma\,(s \cdot (I_R(a, c) + b_0))) \tag{15}$$

Table 5: Comparison of fuzzy operators for differentiable reasoning. T-norms/T-conorms evaluated on $(a, b) = (0.9, 0.1)$; Implications evaluated on $(a, c) = (0.9, 0.2)$; Quantifiers evaluated on set $\mathbf{z} = \{0.9, 0.1\}$ for $\forall$ (target=low). *Note: Gradients for T-norms/T-conorms are w.r.t output score; for Implications w.r.t loss.*

Operator	Definition	Score	Gradients	Gradient Characteristics & Learning Implications
***T-norms (Conjunction $\wedge$):** Generalizes AND. Determines credit assignment between conjuncts.*				
Product (Recommended)	$T_P(a, b) = a \cdot b$	0.09	∇_a=0.1 ∇_b=0.9	**Adaptive signal.** $\partial T/\partial a = b$: gradient scales with partner value. Stronger signal to limiting factor; near-zero values attenuate gradients, preventing destabilization.
Gödel	$T_G(a, b) = \min(a, b)$	0.10	∇_a=0 ∇_b=1	**Single-passing.** Gradient flows only to minimum input. Winner-take-all: learning halts for non-minimum inputs, creating vast zero-gradient plateaus.
Łukasiewicz	$T_L(a, b) = \max(0, a+b-1)$	0	∇_a=0 ∇_b=0	**Dead zones.** Gradients vanish when $a+b \leq 1$. Optimization stalls if model not "confident enough"; extremely brittle to initialization.
Yager (p=2)	$\max(0, 1-\Vert\mathbf{1-x}\Vert_p)$	0.10	∇_a=0.10 ∇_b=0.90	**Tunable.** p interpolates: p=1 $\to$ Łukasiewicz, $p\to\infty$ $\to$ Gödel. Euclidean penalty without dead zones.
Hamacher	$\dfrac{ab}{a+b-ab}$	0.10	∇_a=0.09 ∇_b=0.83	**Coupled dynamics.** Each gradient depends on *both* inputs non-linearly. Own truth value modulates sensitivity.
***T-conorms (Disjunction $\vee$):** Generalizes OR. Dual to t-norms: $S(a, b) = 1 - T(1-a, 1-b)$.*				
Prob. Sum (Recommended)	$S_P(a, b) = a+b-ab$	0.91	∇_a=0.9 ∇_b=0.1	**Balanced.** Updates all operands; penalizes redundancy softly. Dual to Product t-norm.
Maximum	$S_G(a, b) = \max(a, b)$	0.90	∇_a=1 ∇_b=0	**Sparse.** Updates only the strongest evidence. Dual to Gödel t-norm.
Bounded Sum	$S_L(a, b) = \min(1, a+b)$	1.0	∇_a=0 ∇_b=0	**Saturation.** Gradients vanish when sum $\geq$ 1. Dual to Łukasiewicz.
***Fuzzy Implications ($\longrightarrow$):** Models rules "if a then c". Loss = $1 - I(a, c)$. Critical for rule-based learning.*				
Goguen (R-impl.)	$I_G(a, c) = \min(1, c/a)$	0.78	$\nabla_a \propto c/a^2$ $\nabla_c \propto 1/a$	**Zero plateau.** Satisfies adjointness but gradient vanishes when satisfied ($a \leq c$). No margin signal.
Reichenbach (S-impl.)	$I_R(a, c) = 1-a+ac$ $\equiv \neg a \vee c$	0.72	∇_a=0.8 ∇_c=−0.9	**Robust.** Loss = $a(1-c)$: symmetric, non-vanishing.
Sigmoidal (Recommended)	$\text{scale}(\sigma(s \cdot (I+b_0)))$ scale maps σ range to $[0, 1]$	—	Boundary-focused	**Corner fix** [58]. Concentrates gradient on decision boundary; dampens extremes.
***Quantifier Aggregation ($\forall, \exists$):** Pools evidence $\mathbf{z}$ from neighborhood. Evaluated on $\mathbf{z} = \{0.9, 0.1\}$ for $\forall$ (target=low).*				
Strict (Min/Max)	$\min(\mathbf{z})$ for $\forall$ $\max(\mathbf{z})$ for $\exists$	0.10	$\nabla_{\min}$=1 others 0	**Sparse.** Gradient flows only to "bottleneck" neighbor.
Mean (Linear)	$\frac{1}{k}\sum z_i$	0.50	∇_i=$1/k$ for all i	**Dilution.** Outliers washed out by majority.
Gen. Mean (Recommended)	$\mathcal{A}_p^\forall(\mathbf{z})$ (Equation 2) $\mathcal{A}_p^\exists(\mathbf{z})$ (Equation 1)	0.36 (p=2)	Weighted by error	**Tunable Focus.** p controls sensitivity to extremes. $p \approx 2$ offers optimal trade-off.

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function. The function $\text{scale}(\cdot)$ *linearly maps the range $[\sigma(s \cdot b_0), \sigma(s \cdot (1 + b_0))]$ to the unit interval $[0, 1]$ to ensure strict boundary satisfaction:*

$$\text{scale}(y) = \frac{y - \sigma(s \cdot b_0)}{\sigma(s \cdot (1 + b_0)) - \sigma(s \cdot b_0)} \tag{16}$$

***Soundness Condition:** Due to the strict monotonicity of $\sigma(\cdot)$ and the linear scaling, minimizing the loss $\mathcal{L} = 1 - I_{\sigma R}(a, c)$ to 0 strictly implies $I_R(a, c) = 1$.*

Definition 6 (Generalized Mean Error Aggregator). *In the main text (Equation 2), we defined the **Truth Aggregator** $\mathcal{A}^\forall(\mathbf{z})$ for the universal quantifier. For the purpose of the loss analysis in proofs, it is convenient to define the corresponding **Error Aggregator** $\mathcal{E}_p^\forall$ over the set* *of instance losses* $\mathbf{l} = \{l_1, \ldots, l_k\}$ *(where* $l_i = 1 - z_i$*):*

$$\mathcal{E}_p^{\forall}(\mathbf{l}) = \left(\frac{1}{k}\sum_{i=1}^{k} l_i^p\right)^{\frac{1}{p}} \tag{17}$$

Relation to Main Text: The two concepts are duals: $\mathcal{A}^{\forall}(\mathbf{z}) = 1 - \mathcal{E}_p^{\forall}(1 - \mathbf{z})$.

Crucial Property (Strict Zero-Condition): *For any* $p \geq 1$, $\mathcal{E}_p^{\forall}(\mathbf{l}) = 0$ *if and only if* $l_i = 0$ *for all* $i \in \{1, \ldots, k\}$.

A.5 Proof of Soundness (Theorem 1)

Theorem (Zero-Loss Semantic Consistency). *Let* $\mathcal{K}$ *be a knowledge base consisting of a finite set of GF formulas. If a GuardNet model trained on* $\mathcal{K}$ *achieves* $\mathcal{L}_{\text{fidelity}}(\theta) = 0$, *then the learned fuzzy interpretation* $\mathcal{G}_\theta$ *satisfies all axioms on the Core Domain* Δ_{core}. *That is, for every axiom* $\phi \in \mathcal{K}$ *and every grounding* $\bar{c} \in \Delta_{\text{core}}^{|\text{FV}(\phi)|}$, *we have* $[\![\phi]\!]_{\mathcal{G}_\theta, \bar{c}} = 1$.

Proof. The proof proceeds by analyzing the structure of the Fidelity Loss function and the properties of the constituent operators.

Step 1: Decomposition of Fidelity Loss. Recall the definition of the Fidelity Loss from Eq. 10:

$$\mathcal{L}_{\text{fidelity}} = \underbrace{\mathcal{E}_p^{\forall}\left(\{\mathcal{L}_{\text{instance}}(\phi; \bar{c}) \mid \phi \in \mathcal{K}, \bar{c} \in \Delta_{\text{core}}\}\right)}_{\text{Axiom Term}} + \underbrace{\sum \mathcal{L}_{\text{witness}}}_{\text{Witness Term}} \tag{18}$$

Note that here we express the total loss using the error aggregator $\mathcal{E}_p^{\forall}$ defined in Def. A.4, which sums the individual instance losses.

Step 2: Strict Zero-Property. The loss components are non-negative:

- The instance loss $\mathcal{L}_{\text{instance}} \in [0, 1]$ because fuzzy truth values are in $[0, 1]$.
- The aggregator $\mathcal{E}_p^{\forall}$ (Generalized Mean of errors) preserves non-negativity.

Therefore, $\mathcal{L}_{\text{fidelity}} = 0$ implies that every term in the aggregation must be exactly 0. Specifically, using the **Strict Zero-Condition** of the generalized mean:

$$\mathcal{E}_p^{\forall}(\mathbf{L}) = 0 \iff \forall l \in \mathbf{L}, l = 0 \tag{19}$$

This implies that for every axiom $\phi \in \mathcal{K}$ and every witness tuple $\bar{c} \in \Delta_{\text{core}}^{|\text{FV}(\phi)|}$:

$$\mathcal{L}_{\text{instance}}(\phi; \bar{c}) = 0 \tag{20}$$

Step 3: From Loss to Truth Value. The instance loss is defined as $\mathcal{L}_{\text{instance}} = 1 - [\![\phi]\!]_{\mathcal{G}_\theta, \bar{c}}$ (Equation 7). Consequently:

$$1 - [\![\phi]\!]_{\mathcal{G}_\theta, \bar{c}} = 0 \implies [\![\phi]\!]_{\mathcal{G}_\theta, \bar{c}} = 1 \tag{21}$$

Step 4: Satisfaction of Logical Structure. We must verify that $[\![\phi]\!] = 1$ implies the logical satisfaction of the formula structure. Let $\phi = \forall \bar{x}(\alpha \to \psi)$. The truth value is calculated via the Sigmoidal Reichenbach implication $I_{\sigma R}$. From Def. A.4, $I_{\sigma R} = 1$ implies (via the inverse of the scaling and sigmoid functions) that the underlying Reichenbach score $I_R([\![\alpha]\!], [\![\psi]\!]) = 1$.

- $I_R(a, b) = 1 - a + ab$.
- $1 - a + ab = 1 \implies a(1 - b) = 0$.
- This equation holds if $a = 0$ (premise false, vacuous truth) or $b = 1$ (conclusion true).

This aligns exactly with the classical logic definition of material implication ($P \to Q \equiv \neg P \lor Q$).

Conclusion. Since $\mathcal{L}_{\text{fidelity}} = 0$ forces the instance loss to be 0 for all axioms grounded over all tuples in the Core Domain Δ_{core}, it follows that the learned interpretation $\mathcal{G}_\theta$ assigns a truth value of 1 to all such groundings. Thus, the model satisfies the knowledge base $\mathcal{K}$ on Δ_{core}. ◻

A.6 Proof of Approximate Satisfaction (Corollary 1)

Proof of Corollary 1. **(i)** By Jensen's inequality applied to the convex function $f(x) = x^p$ for $p \geq 1$,

$$\left(\frac{1}{k_i}\sum_c z_{i,c}\right)^p \;\leq\; \frac{1}{k_i}\sum_c z_{i,c}^p \;\leq\; \varepsilon_i^p.$$

Taking p-th roots yields $\frac{1}{k_i}\sum_c z_{i,c} \leq \varepsilon_i$. The Jensen step is tight when all instance violations are identical.

(ii) Since

$$\max_c z_{i,c}^p \;\leq\; \sum_c z_{i,c}^p \;\leq\; k_i\, \varepsilon_i^p,$$

taking p-th roots gives $\max_c z_{i,c} \leq k_i^{1/p}\, \varepsilon_i$. Combining with $z_{i,c} \leq 1$ yields $\max_c z_{i,c} \leq \min\{1,\; k_i^{1/p}\, \varepsilon_i\}$. The uncapped L_p-to-L_∞ conversion is attained when all dissatisfaction is concentrated on a single grounding. ◻

A.7 Proof of Training Complexity (Theorem 2)

Theorem (**Training Complexity**). *Let $\mathcal{K}$ be a knowledge base with m guarded axioms $\{\forall \bar{x}_i(\alpha_i \rightarrow \psi_i)\}_{i=1}^m$. Let d be the embedding dim, L the MLP layers, B the batch size. A single training iteration has:*

- ***Time Complexity:*** $O\left(m \cdot |E|_{\max} \cdot L \cdot d^2\right)$
- ***Space Complexity:*** $O\left(|\Delta_{\text{core}}| \cdot d + |\mathcal{P}| \cdot L \cdot d^2 + B \cdot |E|_{\max}\right)$

where $|E|_{\max} = \max_i |\text{ext}(\alpha_i)|$.

Proof. We analyze the computational graph constructed by GuardNet for a single training step (forward and backward pass).

A.7.1 Time Complexity Analysis. Consider the processing of a single guarded axiom $\phi = \forall \bar{x}(\alpha(\bar{x}) \rightarrow \psi(\bar{x}))$.

1. **Guarded Indexing (Lookup):** GuardNet first retrieves the valid variable bindings (tuples) that satisfy the guard atom α. This corresponds to querying the ABox for the extension $\text{ext}(\alpha)$. The number of such tuples is $|\text{ext}(\alpha)|$. This is a sparse lookup operation (gathering indices). Cost: $O(|\text{ext}(\alpha)|)$.

2. **Neural Grounding (Feature Construction):** For each tuple $\bar{a} \in \text{ext}(\alpha)$, we retrieve the embeddings of the constants. If $\bar{a} = (c_1, \dots, c_k)$, we fetch $\mathbf{e}_{c_1}, \dots, \mathbf{e}_{c_k}$. Cost: $O(|\text{ext}(\alpha)| \cdot k \cdot d)$. Since k (arity) is small and constant, this is $O(|\text{ext}(\alpha)| \cdot d)$.

3. **Predicate Evaluation (MLP Forward Pass):** The truth value of the body ψ involves evaluating predicate MLPs. Let's assume ψ contains constant number of atoms. For each atom, we pass the concatenated embeddings through an MLP. * Input dimension: $k \cdot d$. * Hidden dimension: d (for simplicity). * Number of layers: L. * Matrix-Vector Multiplication cost per layer: $\mathbf{Wh}$. $\mathbf{W} \in \mathbb{R}^{d \times d}, \mathbf{h} \in \mathbb{R}^d$. Cost is d^2. * Total MLP cost per tuple: $O(L \cdot d^2)$. * Total cost for all guard-satisfying tuples: $O(|\text{ext}(\alpha)| \cdot L \cdot d^2)$.

4. **Aggregation:** We aggregate the $|\text{ext}(\alpha)|$ truth values using the p-mean. This is a linear reduction. Cost: $O(|\text{ext}(\alpha)|)$.

Total Time Complexity: Summing over m axioms in the knowledge base (or a batch of axioms):

$$T_{\text{total}} = \sum_{i=1}^{m} O(|\text{ext}(\alpha_i)| \cdot L \cdot d^2) \tag{22}$$

Let $|E|_{\max} = \max_i |\text{ext}(\alpha_i)|$. Then:

$$T_{\text{total}} \approx O(m \cdot |E|_{\max} \cdot L \cdot d^2) \tag{23}$$

In sparse graphs, $|E|_{\max} \approx O(N)$ (linear in entities) or $O(|\mathcal{E}|)$ (linear in edges), whereas unguarded approaches require $O(N^2)$ or $O(N^k)$.

A.7.2 Space Complexity Analysis. We analyze the memory footprint required to store parameters and intermediate activation maps for backpropagation.

1. **Model Parameters:** * **Entity Embeddings:** We store embeddings for all constants in the Core Domain. Cost: $O(|\Delta_{\text{core}}| \cdot d)$. * **Predicate Networks:** For each of the $|\mathcal{P}|$ predicates, we store an MLP with L layers of size $d \times d$. Cost: $O(|\mathcal{P}| \cdot L \cdot d^2)$.

2. **Intermediate Activations (The Bottleneck):** To compute gradients, we must store the activations for every node in the computational graph *for the current batch*. * Unlike unguarded approaches that instantiate a tensor of size N^k (all possible tuples), GuardNet only instantiates the computational graph for valid guard extensions. * For a batch of axioms B (or processing the whole KB if full-batch), the number of active computation nodes is proportional to the number of satisfying tuples. * Number of tuples $\leq B \cdot |E|_{\max}$. * For each tuple, we store activations of size $O(L \cdot d)$. * Cost: $O(B \cdot |E|_{\max} \cdot L \cdot d)$. Assuming L, d are constants relative to graph size, the scaling factor is $B \cdot |E|_{\max}$.

Total Space Complexity:

$$S_{\text{total}} = \underbrace{O(|\Delta_{\text{core}}| \cdot d)}_{\text{Embeddings}} + \underbrace{O(|\mathcal{P}| \cdot L \cdot d^2)}_{\text{Weights}} + \underbrace{O(B \cdot |E|_{\max})}_{\text{Activations}} \tag{24}$$

Comparison: * **Unguarded (LTN/Neural LP):** The activation term is $O(N^2)$ (for binary predicates), which dominates and causes OOM. * **GuardNet:** The activation term is $O(|E|_{\max})$. For sparse graphs, $|E| \ll N^2$, often $|E| \approx cN$ (linear).

Thus, the complexity is linear w.r.t the number of edges/entities in sparse KGs. □

B Detailed Experimental Setup

B.1 Dataset Statistics

Tab. 6 summarizes the detailed statistics of the benchmarks used. Note that for SNOMED CT and GO, we include the number of TBox axioms converted to GF formulas.

B.2 Hyperparameter Configuration

We determined optimal hyperparameters via grid search based on validation MRR. Tab. 7 details the search space and best configurations for GuardNet.

Table 6: Detailed statistics of knowledge bases. $|\mathcal{C}|$: Constants, $|\mathcal{P}|$: Predicates, $|\mathcal{A}|$: ABox Facts, $|\mathcal{T}|$: TBox Axioms.

Dataset	$\|\mathcal{C}\|$	$\|\mathcal{P}\|$	$\|\mathcal{A}\|$	$\|\mathcal{T}\|$	Avg. Deg.
SNOMED CT	377,280	64	0	391,204	2.1
Gene Ontology	44,562	14	0	186,013	8.4
Yeast PPI	6,438	8	110,482	14,202	34.3
Human PPI	17,620	12	75,302	28,440	8.5

Table 7: Hyperparameter search space and best settings.

Hyperparameter	Search Space	Best (SNOMED)
Embedding Dim (d)	$\{100, 200, 400\}$	200
Learning Rate	$\{1e^{-3}, 5e^{-4}, 2e^{-4}, 1e^{-4}\}$	$2e^{-4}$
Batch Size	$\{256, 512, 1024, 2048\}$	512
Loss Weight λ	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	0.7
Confidence γ	$\{0.0, 0.01, 0.05, 0.1\}$	0.01
Aggregator p	$\{1, 2, 4, 8\}$	2

B.3 Core Domain Growth

Tab. 8 reports the final size of the Guarded Instance Witness Set $|\Delta_G|$ at training convergence. While Δ_H (Herbrand constants) and Δ_A (atom witnesses, bounded by $2|\mathcal{P}|$) are fixed before training, Δ_G grows dynamically as the model's confidence in guard predicates increases. The growth is convergence-coupled: new witnesses require a guard premise above the confidence threshold, so creation decelerates as the fidelity loss converges.

Table 8: Core domain composition at training convergence.

Dataset	$\|\Delta_H\|$	$\|\mathcal{P}\|$	$\|\Delta_A\|$	$\|\Delta_G\|$	$\|\Delta_G\|/\|\Delta_H\|$
Yeast PPI	6,438	8	16	≈12,400	193%
Human PPI	17,620	12	24	≈22,800	129%

The ratio $|\Delta_G|/|\Delta_H|$ indicates how many additional witness constants the model created relative to the original Herbrand universe. The higher ratio for Yeast PPI reflects its denser topology (average degree 34.3 vs. 8.5 for Human PPI), which triggers more guard predicates above the confidence threshold. Despite this growth, total memory remains modest: $|\Delta_{\text{core}}| \cdot d = (|\Delta_H| + |\Delta_A| + |\Delta_G|) \cdot 200 \approx 9.9$ MB for Yeast and 18.2 MB for Human PPI.

C Logic Translation Standards

Table 9: Translation from Description Logic ($\mathcal{DL}$) to Guarded Fragment (GF). Note that A, B are atomic concepts (unary predicates) and R is a role (binary predicate).

Axiom Type	DL Syntax	Guarded Translation (GF)
Subsumption	$A \sqsubseteq B$	$\forall x\,(A(x) \rightarrow B(x))$
Intersection	$A \sqcap B \sqsubseteq C$	$\forall x\,(A(x) \land B(x) \rightarrow C(x))$
Existential (Right)	$A \sqsubseteq \exists R.B$	$\forall x\,(A(x) \rightarrow \exists y\,(R(x,y) \land B(y)))$
Existential (Left)	$\exists R.A \sqsubseteq B$	$\forall x\,(\exists y(R(x,y) \land A(y)) \rightarrow B(x))$
Equivalent Guarded Form (Left): $\forall x, y\,(R(x,y) \land A(y) \rightarrow B(x))$		

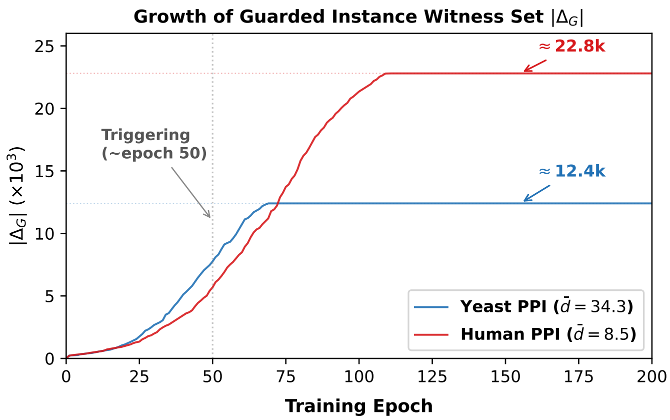

Figure 9: Growth of $|\Delta_G|$ during training. Both curves exhibit logistic growth consistent with the convergence-coupled creation mechanism: witnesses accumulate rapidly once guard confidence rises (~epoch 50), then plateau as fidelity loss converges. Yeast PPI (denser, $\bar{d}$=34.3) triggers witnesses earlier but saturates at a lower count; Human PPI (sparser, $\bar{d}$=8.5) grows later but reaches a higher plateau due to its larger entity set.

D Parameter Sensitivity Analysis

To verify the robustness of GuardNet and justify our configuration choices, we conduct a sensitivity analysis on two critical hyperparameters: the embedding dimension d and the generalized mean exponent p. Fig. 10 visualizes these trends across all four benchmarks, revealing distinct behaviors rooted in data characteristics.

Embedding Dimension (d). As shown in Fig. 10(a), the saturation point of model performance correlates strongly with the semantic complexity of the dataset. For noisy, shallow graphs like Yeast and Human PPI, performance saturates early ($d \approx 100$), as the structural information is relatively sparse. In contrast, deep ontological hierarchies like SNOMED CT and GO exhibit a "long-tail" improvement that persists beyond $d = 200$. This confirms that capturing complex logical subsumption chains requires larger representational capacity. We adopt $d = 200$ as a global equilibrium, sacrificing negligible performance on ontologies for significant memory efficiency.

Aggregator Exponent (p). Fig. 10(b) validates the theoretical properties of the Generalized Mean Error aggregator. While $p = 2$ emerges as the consistent optimum, the sensitivity profiles differ markedly. Strict taxonomic reasoning (SNOMED, GO) demands high logical rigor, resulting in a sharp performance drop at $p = 1$ (arithmetic mean) where logical violations are diluted by satisfied neighbors. Conversely, the noisy topology of PPI networks benefits from the smoothing effect of averaging, manifesting as a broader tolerance curve for lower p values.

E Algorithm Pseudocode

Algorithm 1 outlines the training epoch of GuardNet, highlighting the efficient guarded grounding mechanism.

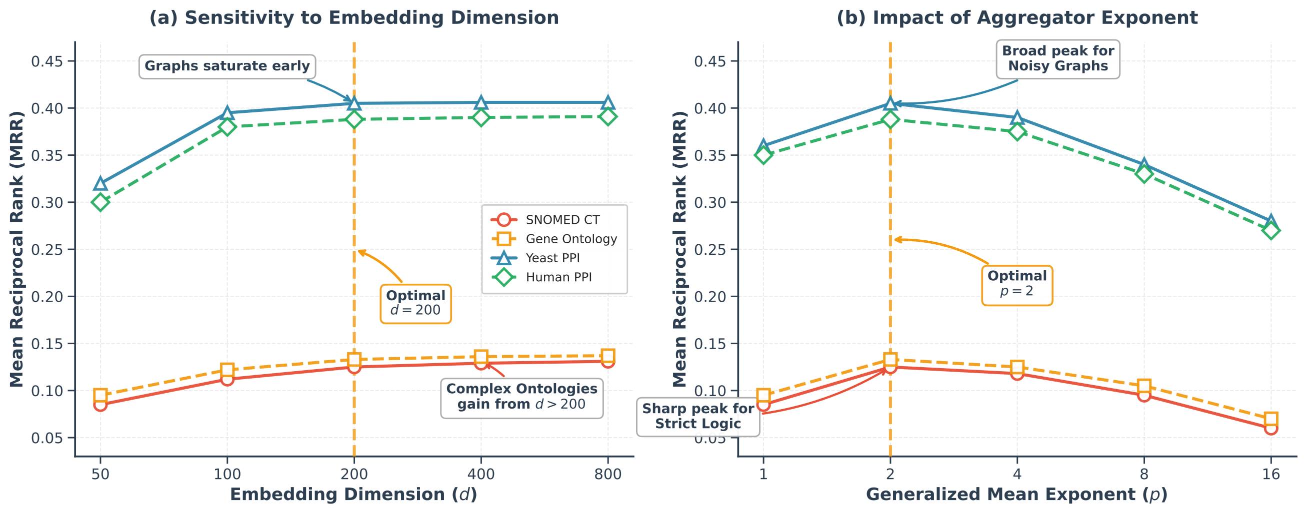

Figure 10: Hyperparameter Robustness across Datasets. (a) Sensitivity to Dimension d: We observe distinct saturation behaviors reflecting data complexity. While noisy PPI graphs saturate early ($d \approx 100$), complex ontologies (SNOMED, GO) show marginal gains beyond $d = 200$, needing capacity to encode deep hierarchies. We select $d = 200$ for global efficiency. (b) Impact of Aggregator p: The optimal $p = 2$ is consistent, though sensitivity varies. Strict taxonomies exhibit a sharp peak (degrading at $p = 1$ as averaging dilutes strictness), whereas noisy PPI graphs show a broader peak (tolerating lower p via smoothing).

Algorithm 1: GuardNet Training Step

Input: Knowledge Base $\mathcal{K}$, Batch Size B, Core Domain Δ_{core}
Output: Updated Parameters θ

for *each batch of axioms* $\{\phi_i\}_{i=1}^{B} \in \mathcal{K}$ **do**
  $\mathcal{L}_{batch} \leftarrow 0$;
  foreach *axiom* $\phi = \forall \bar{x}(\alpha \rightarrow \psi)$ **do**
    `// 1. Guarded Indexing (Topology)`
    $I_{guard} \leftarrow \text{QueryIndex}(\text{ext}(\alpha))$;
    `// 2. Hybrid Domain Sampling`
    $\mathbf{E}_{core} \leftarrow \text{Embed}(\Delta_{core}[I_{guard}])$;
    $\mathbf{E}_{latent} \leftarrow \mathcal{N}(0, I)$; `// Random Latent Samples`
    `// 3. Compute Instance Losses`
    $\mathcal{L}_{core} \leftarrow 1 - I_{\sigma R}(\text{Forward}(\phi, \mathbf{E}_{core}))$;
    $\mathcal{L}_{gen} \leftarrow 1 - I_{\sigma R}(\text{Forward}(\phi, \mathbf{E}_{latent}))$;
    `// 4. Aggregate via Error Aggregator`
    $\mathcal{L}_{batch} \leftarrow \mathcal{L}_{batch} + \lambda \mathcal{E}_p^{\forall}(\mathcal{L}_{core}) + (1-\lambda)\mathcal{E}_p^{\forall}(\mathcal{L}_{gen})$;
  end
  Backward($\mathcal{L}_{batch}$);
  Update(θ);
end

Revision Letter for Paper #271

From Global Search to Local Lookup: Scalable Knowledge Base Completion via Differentiable Guarded Logic

All modifications in the revised paper are highlighted in green. The code repository has been moved to https://anonymous.4open.science/r/guardnet/ to resolve the access issues noted by R2 and R3. We provide a summary of all changes below, followed by detailed responses to the meta-review and each reviewer.

Summary of All Changes

Issue	Change	Location
MR-1	Guard-based Tractability comparison (LTN, TensorLog, GTGDs, Vadalog)	§1
MR-2	Proposition 1, Corollary 1, Theorem 1 renamed	§4.1, §4.4
MR-3	DB-theoretic framing throughout	§1, §2, §6
R1-O1	Novelty survey + 3-axis differentiation	§1
R1-O3	Computational vs. Semantic paragraph	§4.1
R1-O4	Approximate satisfaction corollary + proof	§4.4, App. A
R1-O5	GTGDs/Vadalog venue-fit argument	§1, §2, §6
R2-O1	Hub-node worst-case analysis	§4.1
R2-O2	(s, b_0) sensitivity heatmap + line plot	§5.3, Fig. 8
R2-O3	GF coverage paragraph (4/391K role chains)	§5
R2-O5	$\lvert\Delta_G\rvert$ table + growth curve	App. B
R3-O1	Running example, 9-row operator table, worked example, SNOMED CT terminology, redrawn Figures 3–4	§2–4, App. A
R3-O2	Embedding discussion + FPE ablation	§4.1, Tab. 3

Response to Meta-Review

The meta-reviewer identified three core requirements for the revision. We address each below.

MR-1: Sharper differentiation from prior restricted-quantification NeSy approaches

The revised paper adds a new **Guard-based Tractability** paragraph in §1 that systematically compares three guard mechanisms:

(1) **LTN's guarded quantifier** (Badreddine et al. 2022): a *runtime Boolean mask* applied after materializing the full N^2 domain. It provides no asymptotic improvement over unguarded grounding.

(2) **GTGDs** (Calì et al. 2013): a *syntactic guard condition* that confines the chase to existing tuples, achieving PTIME data complexity for atomic queries.

(3) **Vadalog** (Bellomarini et al. 2018): operationalizes Warded Datalog$^{\pm}$ (strictly more expressive than Guarded Datalog$^{\pm}$) for enterprise-scale reasoning.

GuardNet is positioned as the first system to embed this guard-based tractability mechanism as a *differentiable inductive bias*, connecting NeSy grounding to the DB-theoretic tractability of GTGDs. The paragraph also acknowledges TensorLog (Cohen 2020) as a prior fragment-based approach (polytree-limited SDKGs for inference tractability) and differentiates on bottleneck (inference vs. grounding), fragment (polytree vs. GF), and DB-theoretic lineage. See the detailed discussion under R1-O1 below.

MR-2: Clearer articulation of what is fundamentally novel

We have strengthened the theoretical narrative in three ways:

(1) **New Proposition 1** (§4.1): *Soundness of masked guarded evaluation.* This makes explicit that, under masked guard-index semantics, restricting evaluation to the guard extension is violation-equivalent to full-domain evaluation. An open-world Remark clarifies that the closed-world guard index is a computational convention, not an ontological claim.

(2) **Theorem 1 renamed** to "Zero-Loss Semantic Consistency" (§4.4) with transition text clarifying its role as a consistency guarantee rather than the main theoretical contribution.

(3) **New Corollary 1** (§4.4): *Approximate satisfaction.* For per-axiom error ε_i, the average instance dissatisfaction is bounded by ε_i (tight, via Jensen), and the worst-case violation is bounded by $\min\{1, k_i^{1/p}\varepsilon_i\}$ (sharp). This directly addresses the concern that Theorem 1 was vacuous for nonzero loss.

These results, together with Theorem 2 (Training Complexity), form a three-layer theoretical structure: *sparse evaluation soundness* → *approximate satisfaction* → *computational tractability.*

MR-3: Stronger case for SIGMOD venue fit

The DB-theoretic framing is now woven throughout the paper:

- §1: Guard-based Tractability paragraph explicitly connects to GTGDs and Vadalog.
- §1 (Our Insight): "Building on the GTGD tractability framework..."
- §1 (Contributions): "This neuralizes the tractability mechanism of GTGDs, establishing a formal bridge between database-theoretic query answering and differentiable neural reasoning."
- §2: GF's guard condition identified as the syntactic constraint underlying GTGDs.
- §6 (Conclusion): "This approach neuralizes the tractability mechanism of GTGDs from database theory."

The core message is that GuardNet's guard atom functions as a *differentiable database index*, converting the nested-loop join of unguarded grounding into an index-only lookup. This positions the contribution squarely within the SIGMOD tradition of tractability through structural restrictions.

Presentation accessibility: The meta-review also noted that the presentation could be more accessible. We have substantially revised the exposition following R3's concrete suggestions: see R3-O1 below for details (running example, worked example, table restructuring, SNOMED CT terminology throughout).

Response to Reviewer 1

We thank R1 for upgrading the evaluation and for the constructive feedback. The two required changes (O1, O5) are addressed below; we additionally address O3 and O4. Regarding baseline fairness/completeness (M1): as clarified in our rebuttal, the H@1 metric was reported for all datasets in Table 2; R1 confirmed this concern was resolved.

[R1-O1] Novelty differentiation (Required)

"The operational mechanism—restricting quantifier instantiations to tuples supported by a relational predicate—is conceptually close to common restrictions already used in neuro-symbolic systems (e.g., quantification over observed tuples / typed or sorted domains / relation-scoped grounding)." —R1

We fully agree that restricting quantifier scope is a widespread NeSy practice. GuardNet's novelty is not in the restriction itself, but in the *identification*: recognizing that observed-tuple grounding is an instance of the Guarded Fragment, and that this recognition unlocks a provable $O(|\mathcal{E}|)$ grounding-size bound inherited from the DB-theoretic tractability of GTGDs. Just as Calì et al. proved that the guarded chase terminates in PTIME by confining variable bindings to existing tuples, GuardNet confines grounding to the guard extension for the same structural reason. No prior NeSy system makes this connection. Identifying the fragment is not merely labeling an existing practice; it is what enables the provable bound: without the GF identification, one cannot derive the $O(|\mathcal{E}|)$ complexity from first principles—one can only observe empirical sparsity.

To substantiate this claim, the revised paper now explicitly surveys the representative systems (§1, Guard-based Tractability). We examined TensorLog, Neural-LP, NTP, PSL/NeuPSL, pLogicNet, and NBFNet and confirm three recurring heuristics: observed-tuple restriction, typed domains, and relation-scoped grounding. Among these, TensorLog is the closest prior work: it identifies polytree-limited SDKGs as a tractable fragment with a formal linear-time inference theorem.

GuardNet differs from *all* surveyed systems on three axes:

(1) **Different bottleneck.** TensorLog's polytree restriction targets inference complexity (#P → polynomial); GuardNet's GF guard targets *grounding size* ($O(N^2) \to O(|\mathcal{E}|)$). These are complementary, not redundant.

(2) **Provable bound from a named fragment.** No prior NeSy system we surveyed provides a provable grounding-size bound derived from a named logical fragment. LTN's "guarded quantifier" is a runtime Boolean mask unrelated to the Andréka–van Benthem–Németi Guarded Fragment; Neural-LP/DRUM inherit TensorLog's sparse-matrix sparsity pattern without a fragment-level guarantee; NTP/pLogicNet rely on pruning heuristics.

(3) **DB-theoretic lineage.** GuardNet is the first NeSy system to connect grounding tractability to GTGDs and the Vadalog line of work, establishing a formal bridge between database-theoretic query answering and differentiable neural reasoning.

Where: §1, "The idea of restricting a logical language…"; §1, "Building on the GTGD tractability framework…"; §2, "the guard condition in Definition 1…"

[R1-O3] Classical GF vs. fuzzy regime (Additionally addressed)

"The manuscript should be more careful in how it uses classical GF theory to support claims of principledness. Claims implying that classical GF guarantees transfer directly to the differentiable fuzzy setting are not convincingly established." —R1

The revised §4.1 now contains a "Computational vs. Semantic Properties" paragraph that makes this distinction explicit: the $O(|\mathcal{E}|)$ bound is a *syntactic/topological* property of the guard atom, independent of fuzzy operator choice. Classical semantic properties (decidability, finite

model property) do not automatically transfer to the fuzzy relaxation; the complexity bound does, because it is determined before any truth-value computation begins. A new Proposition 1 (Soundness of masked guarded evaluation) formalizes the sparse evaluation semantics, with an open-world Remark clarifying the computational convention.
Where: §4.1, "The $O(|\mathcal{E}|)$ bound is a computational property…"; Proposition 1, "For sparse grounding, GuardNet uses…"; Remark, "The closed-world guard index…"

[R1-O4] Theorem 1 vacuous (Additionally addressed)

"Theorem 1 is essentially vacuous for realistic KBC…A more useful theoretical contribution would discuss approximate satisfaction." —R1

We have restructured the theoretical narrative:

- Theorem 1 renamed "Zero-Loss Semantic Consistency" with transition text clarifying it as a consistency guarantee, not the main contribution.
- New Corollary 1 (Approximate satisfaction): for per-axiom error ε_i, bounds average dissatisfaction ($\leq \varepsilon_i$, tight) and worst-case violation ($\leq \min\{1, k_i^{1/p}\varepsilon_i\}$, sharp). Proof in Appendix A.
- Theorem 2 (Training Complexity) foregrounded as the main scalability guarantee.

Where: §4.4, Theorem 1 "Zero-Loss Semantic Consistency"; "Theorem 1 establishes that the loss function…"; Corollary 1 "Approximate satisfaction"; Appendix A (proof).

[R1-O5] SIGMOD venue fit (Required)

"The paper is still primarily a neuro-symbolic learning method…without a concrete database/graph-system artifact." —R1

Addressed jointly with MR-3 above. The DB-theoretic framing (GTGDs, Vadalog, guard atom as differentiable index) is now present in §1, §2, and §6. The core argument: GuardNet's grounding mechanism is the differentiable analogue of the guarded chase in database theory, and its $O(|\mathcal{E}|)$ bound inherits the PTIME data complexity of GTGDs for atomic queries.

Response to Reviewer 2

We thank R2 for the detailed technical scrutiny. All four issues are addressed.

[R2-O1/O4] Hub-node density (Required)

"High-degree hub nodes might temporarily approximate $O(N^2)$ local density during mini-batching…worst-case memory spikes for highly skewed, scale-free graphs." —R2

The revised §4.1 (Sparsity Assumption) adds a third caveat addressing scale-free networks. In our densest experimental dataset (N=6,438, $\bar{d}$=34.3), power-law degree analysis estimates $d_{\max} \approx 300$, so even the largest hub processes $d_{\max}^2/N^2 < 0.3\%$ of the pairs an unguarded approach would evaluate. Hub load imbalance is a scheduling concern handled by our mini-batch strategy, not an asymptotic one.
Where: §4.1, "Even in scale-free networks with high-degree hub nodes…"

[R2-O2] (s, b_0) sensitivity (Required)

"Sensitivity of the guarded instance witness generation (Δ_G) to specific scalar thresholds is unclear." —R2

We added a (s, b_0) sensitivity analysis with both a heatmap and line plot as Figure 8 in §5.3 (promoted from Appendix to main text). MRR shifts by at most 1.5% across a 4× range of steepness s, confirming robustness. Performance is stable across bias values $b_0 \in [-0.7, -0.3]$.
Where: §5.3, Figure 8.

[R2-O3] GF restriction as downsampling; non-guarded rules (Required)

"Restricting computation to guarded rules acts in a similar approach as majority-class downsampling…reasoning degrades when encountering necessary non-guarded rules outside the GF." —R2

The revised §5 adds a GF Coverage paragraph. The downsampling analogy does not hold: unguarded tuples contribute *vacuous truths* (zero semantic content, zero gradient), not discriminative signal. Coverage analysis confirms that all $\mathcal{EL}^{++}$ constructs except role chains and transitivity are expressible in GF. These exceptions are negligible: SNOMED CT contains only 4 such axioms out of 391K total. For excluded axioms, GuardNet applies bounded-depth neighborhood sampling as a fallback.
Where: §5, "The only $\mathcal{EL}^{++}$ constructs that fall outside GF…"

[R2-O5] $|\Delta_G|$ growth (Required)

"Scaling of Δ_G across epochs is not clearly addressed. If many rules trigger dynamically, the active domain size could inflate substantially." —R2

Appendix B now contains both a convergence table ($|\Delta_G|/|\Delta_H|$ ratios and memory footprint) and a new growth curve figure. Both curves exhibit logistic growth: witnesses accumulate rapidly once guard confidence rises (~epoch 50), then plateau as fidelity loss converges. Peak $|\Delta_G|$ reaches ≈12.4K (Yeast) and ≈22.8K (Human), with total embedding memory ≤18.2 MB.
Where: Appendix B, Table and growth curve figure.

Response to Reviewer 3

We thank R3 for the positive assessment and the concrete presentation suggestions. All changes are implemented.

[R3-O1] Presentation: Definition 2, Table 1, examples (Required)

"I don't see there being much value to defining GF's semantics in the rigorous way of Definition 2…Is Table 1 necessary?…Provide the reader with some concrete examples!" —R3

(1) **Definition 2** replaced by an intuitive running example (Example 1: two SNOMED CT axioms with display-line translations). The formal fuzzy satisfaction definition moves to Appendix A.
(2) **Table 1** moves to Appendix A. A 9-row summary table remains in §3, listing the selected operators (bold) and all evaluated alternatives with gradient annotations. We expanded beyond the originally promised 5 rows to cover all operators appearing in the heatmap analysis (Figure 7), ensuring the main-text table is self-contained for interpreting the experimental results.
(3) **Worked example** added as a Guarded Grounding Trace box in §4.1, walking through a 3-entity SNOMED CT subgraph. The Cat/Tail illustrative example throughout §4.2–4.3 has been fully replaced with Pneumonia/Fever/hasFinding terminology consistent with Example 1. Figures 3–4 have been redrawn with matching SNOMED CT entities (Pneumonia, Fever, hasFinding replacing Cat, Tail, hasPart).

Where: §2, "We illustrate with two axioms from SNOMED CT…"; §3, Table 1; §4.1, "Consider a 3-entity subgraph…"; §4.2–4.3 (terminology, Figures 3–4); Appendix A.

[R3-O2/Q1] Learnable embeddings (Required)

"Is 'constants as learnable embeddings' discussed anywhere besides the one paragraph? What role does that play in GuardNet and contribute to rule satisfaction?" —R3

The revised §4.1 expands the embedding discussion: entity and relation embeddings are jointly optimized end-to-end within the fuzzy loss, paralleling how TransE/RotatE learn from link-prediction objectives but with logical axioms as supervision. A new ablation row (Row 6, "FPE" in Table 3) fixes pretrained TransE embeddings without fine-tuning, isolating the contribution of joint optimization. The result confirms that learnable embeddings are essential: FPE degrades MRR by 5–12% across all benchmarks.
Where: §4.1, "entity and relation embeddings are jointly optimized…"; §5, Table 3 Row 6 (FPE).

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